NEM 201
Fundamentals of Nematology

Complete Lecture-wise Theory Study Material

B.Sc. (Hons) Agriculture — Third Semester — TNAU

2025 Syllabus — All 17 Lectures

Exam-oriented notes with tables, worked examples, text diagrams and model answers

How to Use This Material

This book is a complete, lecture-wise theory companion for NEM 201 - Fundamentals of Nematology (B.Sc. (Hons) Agriculture, Third Semester, Tamil Nadu Agricultural University, 2025 syllabus). It follows the official syllabus lecture by lecture, in the exact order in which the topics are taught.

Each lecture is written as a self-contained chapter. Inside every chapter the sub-topics appear in syllabus order, under clear headings, in short exam-oriented sentences. Definitions are given in plain English first, then the technical detail.

What Each Chapter Contains

- Headed sections that expand every sub-topic of the syllabus line, in order.
- At least one comparison table, worked example or labelled text diagram in every lecture.
- Indian and Tamil Nadu examples wherever the syllabus allows - rice, banana, tomato, pulses, coconut, and TNAU-recommended practices such as Pseudomonas fluorescens, Trichoderma viride and Paecilomyces lilacinus.
- An "Exam focus" box at the end of every lecture: key points, three short-answer questions with model answers, and two long-answer questions with answer outlines.

Worked examples are shown step by step so that the arithmetic can be reproduced in an examination. Where a field figure varies with soil, climate or cultivar, a standard published range is given instead of a single false-precise number.

Syllabus Map (All 17 Lectures)

No.	Lecture topic
1	Introduction - brief history and development of nematology at national and international levels
2	Position of nematodes in the animal kingdom; importance (to humans, animals, plants; insect parasites); habitat and diversity; economic loss in crop plants
3	General morphology of nematodes
4	Anatomy of nematodes - digestive, excretory and nervous systems and sense organs
5	Anatomy of nematodes - reproductive system
6	Biology of economically important nematode genera: Meloidogyne, Heterodera, Tylenchulus, Rotylenchulus and Radopholus
7	Taxonomy of plant-parasitic nematodes of the Secernentea and Adenophorea
8	Classification of plant-parasitic nematodes based on feeding habits; beneficial nematodes
9	Mid Semester Examination
10	Symptoms of nematode damage and role of nematodes in disease development
11	Interaction between plant parasitic nematodes and disease-causing fungi, bacteria and viruses
12	Nematode parasites of cereals
13	Nematode parasites of vegetable crops
14	Nematode parasites of pulses, oilseed and fibre crops
15	Nematode parasites of fruit and plantation crops
16	Principles of nematode management (legislative, physical, cultural, biological, chemical and host-plant resistance)

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No.	Lecture topic
17	Integrated nematode management

Lecture 9 is the Mid Semester Examination. No new theory is prescribed for it; the examination is drawn from Lectures 1 to 8, so this book carries a short revision checklist in place of a theory chapter.

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Lecture 1: Introduction - Brief History and Development of Nematology

1.1 What Nematology Is

Nematology is the branch of biological science that deals with nematodes. The word comes from the Greek *nema* (thread) and *logos* (study). Nematodes are therefore literally "thread-like animals". They are also called eelworms, roundworms and threadworms.

A nematode is a soft-bodied, unsegmented, bilaterally symmetrical, worm-like animal that is round in cross-section and lives either free in soil and water or as a parasite of plants, animals and insects. Plant-parasitic nematodes form a small but economically very important part of this group.

> *Nematology is a link science. It borrows from zoology (the animals), plant pathology (the diseases), soil science (the habitat) and agronomy (the management). This is why the subject is taught to agriculture students and not only to zoologists.*

1.2 Why the History Matters

The history of nematology shows how a group of animals that nobody could see with the naked eye grew into a full agricultural science. Three strands run through that history:

- Recognition - people first noticed nematodes because of the damage they caused (galled wheat, sick sugar beet).
- Basic science - the structure, anatomy and classification of nematodes was worked out between about 1870 and 1950.
- Applied science - from about 1950, nematicides, resistant varieties, biological control and integrated management turned nematology into a management discipline.

1.3 Development at the International Level

1.3.1 The early observations (1743-1900)

The first plant-parasitic nematode ever recorded was the wheat seed-gall nematode, now called *Anguina tritici*. In 1743 the English clergyman-scientist Needham examined "blighted" wheat grains and found tiny eel-like worms inside them; he called them *Vibrio tritici*. In 1855 Berkeley named the organism and also recorded galls on the roots of cucumber. In 1857 Kuhn studied the wheat eelworm in detail, and this is often taken as the beginning of plant nematology as a serious subject.

The sugar-beet cyst nematode, *Heterodera schachtii*, was discovered in Germany on sugar beet in 1859 by Schacht, and was formally described by Schmidt in 1871. Root-knot nematodes were first seen in coffee by Goeldi in 1889, who named *Meloidogyne exigua*; for many years, however, all root-knot nematodes were placed in one species and were called *Heterodera marioni*.

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1.3.2 The modern era begins (1900-1950)

The single most important name in nematology is Nathan Augustus Cobb (1913 onwards). Working at the United States Department of Agriculture, Cobb laid the foundations of nematode morphology, described hundreds of genera, developed extraction and counting methods, and coined the very word "nematology". He is rightly called the father of nematology.

In 1933 Chitwood and Chitwood divided the phylum into the two great classes Secernentea and Adenophorea (then called Phasmidia and Aphasmidia), a classification that is still used as a first split. In 1949 Chitwood separated the root-knot nematodes into the genus *Meloidogyne*, which finally allowed the different species to be recognised and studied.

1.3.3 The applied era (1950 to the present)

After the Second World War, cheap and effective soil nematicides (such as DBCP, ethylene dibromide and methyl bromide) became available, and nematology grew rapidly as a management science. Important landmarks of this period include:

- The Society of Nematologists (SON) was founded in the USA in 1961, and the European Society of Nematologists (ESN) in the 1950s-60s; the Organization of Nematologists of Tropical America (ONTA) followed in 1967.
- Research journals appeared: *Nematologica* (1956), the *Journal of Nematology* (1969) and later the *Revue de Nematologie*.
- The International *Meloidogyne* Project (1975-1984), led by J. N. Sasser, surveyed root-knot nematodes across the world and produced the first global picture of the problem.
- The book *Nematology - Fundamentals and Recent Advances* by Sasser and Jenkins (1960) became a standard teaching text.
- In 1987 Sasser and Freckman published a world perspective on nematology and estimated global crop losses of about US $77 billion per year, a figure later revised upward to roughly US $80-125 billion.
- Molecular tools (DNA barcoding, genomics) arrived from the 1990s. The genome of the root-knot nematode *Meloidogyne incognita* was published in 2014, opening the door to molecular breeding and RNA-interference-based control.

Year	Milestone in the development of nematology
1743	Needham records eelworms in blighted wheat grain (Anguina tritici) - first plant-parasitic nematode
1855	Berkeley names the wheat eelworm and records root galls on cucumber
1859-1871	Sugar-beet cyst nematode Heterodera schachtii found (Schacht 1859) and described (Schmidt 1871)
1889	Goeldi names Meloidogyne exigua in coffee - first root-knot record
1913	N. A. Cobb begins his work; the term "nematology" is coined
1933	Chitwood and Chitwood split the phylum into Secernentea and Adenophorea
1949	Chitwood establishes the genus Meloidogyne
1956	Journal Nematologica launched
1960	Sasser and Jenkins publish a standard nematology textbook
1961	Society of Nematologists (USA) founded

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Year	Milestone in the development of nematology
1975-1984	International Meloidogyne Project, led by J. N. Sasser
1987	Sasser and Freckman estimate world crop loss at about US $77 billion per year
2008-2014	Genomes of Meloidogyne hapla (2008) and M. incognita (2014) published

1.4 Development at the National Level (India)

1.4.1 Early reports

Plant-parasitic nematodes were noticed in India well before any organised research began. The root-knot nematode was reported on tea in Assam by Barber in 1901, one of the earliest Indian records. The wheat seed-gall nematode (*Anguina tritici*), known to farmers as "seeds turning into galls", has been a traditional problem of wheat for over a century, and "molya" disease of wheat in Rajasthan (caused by the cereal cyst nematode *Heterodera avenae*) is an old, locally named malady.

1.4.2 Organisation of nematology in India

Systematic work began in the middle of the twentieth century:

- A separate Division of Nematology was established at the Indian Agricultural Research Institute (IARI), New Delhi, in the late 1950s. This was the first dedicated nematology research unit in the country.
- Departments of Nematology were later created in agricultural universities, including the Tamil Nadu Agricultural University (TNAU), Coimbatore, and other state agricultural universities, to teach the subject and to run research and advisory programmes.
- The Nematological Society of India (NSI) was formed around 1969-1971, and the *Indian Journal of Nematology* began publication in 1971.
- The Indian Council of Agricultural Research (ICAR) launched an All India Coordinated Research Project on nematodes, so that a common research programme could be run across states, including Tamil Nadu.
- Today, nematode management is a standard recommendation in crop packages of practice, and TNAU has released specific advisories using the bioagents *Pseudomonas fluorescens*, *Trichoderma viride* and *Paecilomyces lilacinus*.

Period / Year	Development in India
1901	Root-knot nematode reported on tea in Assam (Barber) - one of the earliest Indian records
Late 1950s	Division of Nematology set up at IARI, New Delhi - first dedicated unit in India
1960s onward	Nematology departments and teaching programmes in state agricultural universities, including TNAU
c. 1969-1971	Nematological Society of India formed; Indian Journal of Nematology launched (1971)
1970s onward	ICAR All India Coordinated Research Project on nematodes; coordinated testing across states
Present day	Nematode management built into crop packages; bioagent-based advisories (P. fluorescens, T. viride, P. lilacinus) by TNAU and others

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1.5 Modern Trends in Nematology

- Molecular taxonomy and diagnostics: DNA barcoding now identifies nematode species and populations that look alike under the microscope.
- Genomics and functional biology: complete genomes and gene-expression studies explain how nematodes form feeding cells in roots.
- Host resistance and breeding: resistant/tolerant varieties and marker-assisted selection are being used against root-knot and cyst nematodes.
- Biological control: *Paecilomyces lilacinus*, *Pochonia chlamydosporia*, *Trichoderma* spp., *Pseudomonas fluorescens* and *Pasteuria penetrans* are now standard bioagents.
- Integrated nematode management (INM): the modern approach combines resistant varieties, cultural methods, bioagents and need-based chemicals instead of depending on a single method.
- Nematodes as bioindicators: because different nematodes respond differently to soil disturbance, the nematode community is used as an index of soil health.

1.6 Scope and Branches of Nematology

Because nematodes occur in so many habitats, nematology has grown into several branches.

- Agricultural (plant) nematology - the study of plant-parasitic nematodes and their management; the main branch for agriculture students.
- Medical nematology - the study of nematodes that parasitise human beings (Ascaris, hookworm, filarial worms).
- Veterinary nematology - the study of nematodes of livestock and pets (Haemonchus, Ostertagia, Toxocara).
- Marine and freshwater nematology - the study of free-living aquatic nematodes, which are very abundant and important in aquatic food webs.
- Soil ecology - the study of free-living soil nematodes as part of the soil food web.
- Molecular nematology - the study of nematode genes, genomes and molecular interactions with hosts.
- Insect nematology - the study of entomopathogenic nematodes used in pest control.

1.7 Institutions and Journals

The important international bodies are the Society of Nematologists (SON), the European Society of Nematologists (ESN) and the Organization of Nematologists of Tropical America (ONTA). Important journals include Nematologica (now Nematology), the Journal of Nematology and the Indian Journal of Nematology. In India, work is carried out at the Indian Agricultural Research Institute (IARI), the Indian Council of Agricultural Research (ICAR) network projects, and state agricultural universities including TNAU, Coimbatore.

Exam focus - Lecture 1

Key points. (1) Nematology is the study of thread-like, unsegmented roundworms. (2) The first plant-parasitic nematode recorded was Anguina tritici, seen by Needham in blighted wheat in 1743. (3) Heterodera schachtii (sugar beet, 1859) and Meloidogyne (coffee, 1889; genus established 1949) are the other historic landmarks. (4) N. A. Cobb (1913) is the father of nematology and coined the

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term. (5) Chitwood and Chitwood (1933) divided the phylum into Secernentea and Adenophorea. (6) In India, root-knot was reported on tea in 1901 and the first dedicated unit (IARI Division of Nematology) was set up in the late 1950s. (7) Sasser and Freckman (1987) put world crop loss at about US $77 billion per year. (8) Modern nematology uses molecular diagnostics, genomics, bioagents and integrated management.

Short-answer questions.

Q1. Who is called the father of nematology and why? *Model answer:* N. A. Cobb (1913). He founded nematode morphology and taxonomy, described many genera, developed extraction and counting methods, and coined the term "nematology"; his work turned a scattered subject into a science.

Q2. Name the first plant-parasitic nematode ever recorded and the crop on which it was found. *Model answer:* Anguina tritici (then called Vibrio tritici), found by Needham in 1743 in blighted (galled) wheat grain.

Q3. What was the contribution of Chitwood and Chitwood (1933)? *Model answer:* They divided the phylum Nematoda into two classes - Secernentea (Phasmidia) and Adenophorea (Aphasmidia) - a basic split based mainly on the presence or absence of phasmids and other features.

Long-answer questions.

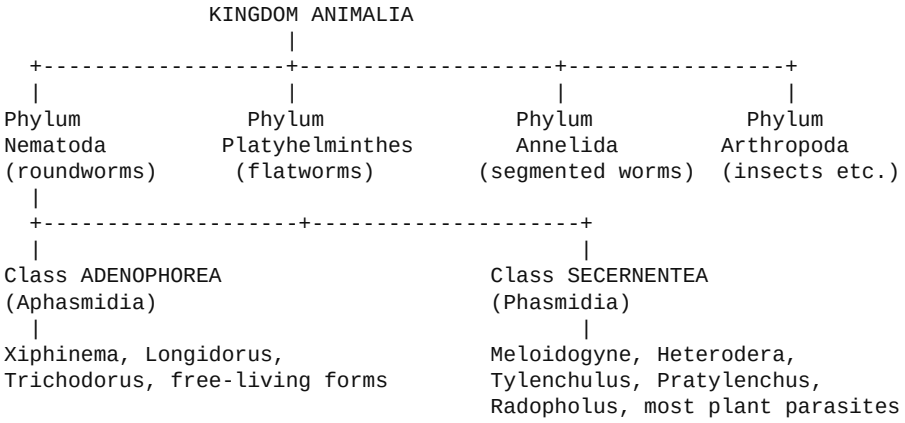
Q1. Trace the development of nematology at the international level. *Outline:* early records (Needham 1743; Berkeley 1855; Heterodera schachtii 1859-71; Meloidogyne 1889) - Cobb and the founding of morphology and taxonomy (1913) - Chitwood and Chitwood classification (1933) - genus Meloidogyne separated (1949) - applied era: nematicides and societies from the 1950s-60s - International Meloidogyne Project (1975-84) - global loss estimates (1987) - molecular and genomic era (2008-2014).

Q2. Describe the growth of nematology in India and its present status. *Outline:* early field reports (root-knot on tea, 1901; molya of wheat) - IARI Division of Nematology in the late 1950s - teaching and research in agricultural universities including TNAU - Nematological Society of India and Indian Journal of Nematology - ICAR coordinated project - today: management in crop packages, bioagent advisories, molecular diagnostics.

Lecture 2: Position of Nematodes in the Animal Kingdom - Importance, Habitat, Diversity and Economic Loss

2.1 Position of Nematodes in the Animal Kingdom

Nematodes form the phylum Nematoda within the kingdom Animalia. They are triploblastic, bilaterally symmetrical, unsegmented, pseudocoelomate animals that are round in cross-section. Older textbooks place them as a class (Nematoda) inside a larger phylum called Nemathelminthes or Aschelminthes; modern zoology treats Nematoda as a full phylum in its own right.



2.2 Diagnostic Features of the Phylum

- Body unsegmented, cylindrical and round in cross-section (hence "roundworm").
- Body covered by a non-living cuticle; no cilia on the general body surface in most species.
- Body cavity is a pseudocoelom, filled with fluid, which acts as a hydrostatic skeleton.
- Only longitudinal body-wall muscles are present, so the body bends dorsoventrally and cannot loop like an earthworm.
- Digestive tract is a complete tube with a mouth and an anus.
- No circulatory and no respiratory system; gas exchange takes place by diffusion through the body surface.
- Sexes are usually separate (dioecious), with clear sexual dimorphism.

Feature	Nematodes (roundworms)	Earthworms (annelids)
Body form	Unsegmented, cylindrical, tapers at both ends	Segmented, cylindrical
Body cavity	Pseudocoelom	True coelom
Muscles	Longitudinal only	Circular and longitudinal
Cilia on surface	Absent	Present in many
Movement	Bends dorsoventrally, wriggling	Crawling by peristalsis
Sexes	Usually separate	Usually hermaphrodite

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2.3 Importance of Nematodes

Nematodes matter in three directions - to human beings, to animals, and to plants - and they are also useful parasites of insects.

2.3.1 Importance to human beings

- As human parasites (harmful): *Ascaris lumbricoides* (roundworm of the gut), hookworms (*Ancylostoma*, *Necator*), *Wuchereria bancrofti* (lymphatic filariasis), *Enterobius vermicularis* (pinworm), *Trichuris trichiura* (whipworm), *Strongyloides stercoralis*, *Trichinella spiralis* (from undercooked pork) and *Onchocerca volvulus* (river blindness). These are major public-health problems in tropical countries, including parts of India.
- As a model organism (useful): the free-living *Caenorhabditis elegans* is one of the most studied animals in biology. Work on it won the Nobel Prize in Physiology or Medicine in 2002 (Brenner, Horvitz and Sulston) and it is central to genetics, development and neurobiology.
- As soil-health indicators (useful): the nematode community in soil is used as a bioindicator, because different nematodes respond differently to farming practices and pollution.
- In nutrient cycling (useful): free-living soil nematodes graze on bacteria and fungi and release plant nutrients, so they help recycle nitrogen.

2.3.2 Importance to animals

- Livestock parasites such as *Haemonchus contortus* (barber's pole worm of sheep and goats), *Ostertagia* and *Cooperia* (cattle), cause serious production losses in grazing animals.
- *Toxocara* in dogs and cats and *Dirofilaria immitis* (heartworm) are important animal nematodes.
- On the useful side, entomopathogenic nematodes (see 2.3.4) are used to control insect pests of crops and are harmless to humans and livestock.

2.3.3 Importance to plants

Plant-parasitic nematodes number about 4,100 described species. Although this is only a small fraction of all nematodes, they are the reason nematology is taught in agriculture. They feed on roots, stems, leaves, bulbs and seeds and reduce both yield and quality. The most important genera are *Meloidogyne* (root-knot), *Heterodera* and *Globodera* (cyst), *Pratylenchus* (lesion), *Radopholus* (burrowing), *Tylenchulus* (citrus), *Rotylenchulus* (reniform), *Xiphinema*, *Longidorus* and *Trichodorus* (ectoparasites and virus vectors), *Ditylenchus* (stem and bulb) and *Aphelenchoides* (foliar).

2.3.4 Parasites of insects

Some nematodes attack insects and are therefore useful in pest control. They are called entomopathogenic nematodes.

- *Steinernema* and *Heterorhabditis* carry symbiotic bacteria (*Xenorhabdus* and *Photorhabdus* respectively). The nematode enters the insect, releases the bacterium, and the insect dies from the bacterial infection within a day or two. Formulations of these nematodes are sold as biopesticides against borers, grubs and other soil insects, and *Heterorhabditis indica* is an Indian isolate used for this purpose.

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- *Deladenus* (*Beddingia*) *siricidicola* sterilises the female of the *Sirex* wood wasp, an important forestry pest.
- *Romanomermis* species parasitise mosquito larvae and have been tested for mosquito control.

Direction	Examples and role
Human health (harmful)	Ascaris, hookworm, Wuchereria (filariasis), Enterobius, Trichuris, Onchocerca
Human science (useful)	Caenorhabditis elegans as a model organism; nematodes as soil-health bioindicators
Animal health (harmful)	Haemonchus, Ostertagia, Cooperia in livestock; Toxocara, Dirofilaria in pets
Plants (harmful)	Meloidogyne, Heterodera, Pratylenchus, Radopholus, Tylenchulus, Rotylenchulus and others
Insect control (useful)	Steinernema, Heterorhabditis (with symbiotic bacteria); Deladenus; Romanomermis

2.4 Habitat and Diversity

2.4.1 Habitat

Nematodes are found almost everywhere that moisture and organic matter exist. Their habitats include:

- Soil - the richest habitat; a single handful of fertile soil can contain several thousand nematodes. Most plant-parasitic and free-living forms live here.
- Freshwater - ponds, rivers, lakes and the water film around soil particles.
- Marine - from the intertidal zone to the deep sea floor; marine nematodes are among the most abundant animals on the seabed.
- Extreme environments - hot springs, Antarctic ice, deserts (in a dehydrated state) and even inside other animals (as parasites).

2.4.2 Feeding diversity

Free-living nematodes occupy every feeding type in the soil food web:

- Bacterivores - feed on bacteria.
- Fungivores - feed on fungi.
- Algivores - feed on algae.
- Omnivores - feed on a mixture of food.
- Predators - attack other nematodes and small soil animals.

2.4.3 How diverse are nematodes?

- About 25,000 species have been described so far, but estimates of the true number range from about 500,000 to more than 1,000,000 species. Nematoda is one of the most species-rich phyla, second only to the arthropods in most estimates.
- Nematodes are extremely abundant. It is commonly said that about four out of every five animals on earth is a nematode.

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- N. A. Cobb famously wrote that if all the matter in the universe except nematodes were swept away, the world would still be dimly recognisable, with the nematodes marking out the mountains, rivers, towns and the places where people had lived.

Diversity is high in form as well as in number: from the microscopic free-living species of a millimetre or less, to the giant Placentonema gigantissima (about 8 m long) found in the placenta of the sperm whale.

2.5 Economic Loss in Crop Plants

2.5.1 Global loss

Nematodes are called the "hidden enemy" because their damage is often mistaken for poor soil, drought or nutrient deficiency. Sasser and Freckman (1987) estimated a world crop loss of about US $77 billion per year; later estimates place the figure in the range of US $80-125 billion per year, roughly 10-14 per cent of global crop production. Losses are usually higher in tropical and subtropical countries, where several nematode species attack the same crop.

2.5.2 Losses in India and Tamil Nadu

No single national figure is agreed, and published estimates vary widely, but Indian crop losses to nematodes are commonly put at several thousand crore rupees per year. Important Indian examples are the cereal cyst nematode ("molya") of wheat in Rajasthan, root-knot in vegetables and pulses, the citrus nematode in citrus orchards, and the burrowing nematode in banana.

Crop / region	Main nematode	Typical loss (standard range)
Wheat, Rajasthan ("molya")	Heterodera avenae	up to 40-60 per cent in severe patches
Upland / rainfed rice	Meloidogyne graminicola	about 20-40 per cent
Tomato and other vegetables	Meloidogyne spp.	about 30-60 per cent (higher in heavy soil)
Banana	Radopholus similis	about 20-50 per cent, with toppling
Citrus	Tylenchulus semipenetrans	about 10-30 per cent (slow decline)
Cotton	Rotylenchulus reniformis	about 10-30 per cent
Groundnut	Meloidogyne arenaria	up to about 40 per cent
All crops (world)	many species	about 10-14 per cent (US $80-125 billion per year)

2.6 Nematodes in the Soil Food Web

Free-living nematodes are central to the soil food web. Bacterivorous and fungivorous nematodes graze on micro-organisms and release nitrogen and other nutrients in plant-available form, so they speed up nutrient cycling. Predatory nematodes feed on other nematodes and small soil animals and help to regulate the community. Because each feeding group responds differently to tillage, fertiliser, organic matter and pollution, the mix of nematode feeding groups is used as a soil-health index - a "bioindicator" of how healthy the soil is.

2.7 Nematodes Compared with Other Plant Pathogens

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Feature	Nematodes compared with fungi and bacteria
Size	Much larger than fungi and bacteria (visible under a low-power microscope)
Nature	Animals, not plants or micro-organisms
Movement	Active, can move short distances in the soil water film
Damage	Mostly underground, on roots; symptoms appear above ground
Detection	Need extraction and counting from soil and roots, not just visual inspection
Spread	Mainly by soil, water, planting material and farm implements
Management	Rarely cured; must be managed by prevention and integration of methods

Exam focus - Lecture 2

Key points. (1) Nematodes form the phylum Nematoda in kingdom Animalia; they are triploblastic, bilaterally symmetrical, unsegmented and pseudocoelomate. (2) They have a cuticle, a pseudocoelom, only longitudinal muscles, and no circulatory or respiratory system. (3) The two classical classes are Adenophorea (Aphasmidia) and Secernentea (Phasmidia). (4) Nematodes are important to humans (parasites and the model organism C. elegans), to animals (livestock parasites), to plants (plant-parasitic nematodes) and as insect parasites (Steinernema, Heterorhabditis). (5) Habitats range from soil and freshwater to the deep sea; feeding types include bacterivores, fungivores, omnivores and predators. (6) About 25,000 species are described; total numbers may exceed a million. (7) World crop loss is about US $80-125 billion per year (10-14 per cent).

Short-answer questions.

Q1. Give the systematic position of nematodes. *Model answer:* Kingdom Animalia, phylum Nematoda; triploblastic, bilaterally symmetrical, unsegmented, pseudocoelomate; divided classically into Adenophorea and Secernentea.

Q2. Name two nematodes useful in insect control and explain briefly. *Model answer:* Steinernema and Heterorhabditis. They carry symbiotic bacteria (Xenorhabdus, Photorhabdus) into the insect; the bacteria kill the insect. They are used as biopesticides.

Q3. Why are nematodes called the "hidden enemy" of crops? *Model answer:* Because their underground damage produces general symptoms (stunting, yellowing, wilting) that are easily mistaken for nutrient deficiency, drought or poor soil, so the real cause is often missed.

Long-answer questions.

Q1. Discuss the economic importance of nematodes, covering humans, animals, plants and insects. *Outline:* human parasites (Ascaris, hookworm, filariasis, etc.) - useful aspects (C. elegans, bioindicators, nutrient cycling) - animal parasites (Haemonchus, Ostertagia, Toxocara) - plant-parasitic nematodes and their genera - insect-parasitic nematodes and their use as biopesticides - summary table.

Q2. Describe the habitat and diversity of nematodes and explain the economic loss they cause. *Outline:* habitats (soil, freshwater, marine, extreme environments) - free-living feeding types - abundance and species estimates (about 25,000 described; up to a million estimated; four of every five animals) - Cobb's remark - global loss US $80-125 billion (10-14 per cent) - Indian examples (molya of wheat, root-knot of vegetables, citrus and banana).

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Lecture 3: General Morphology of Nematodes

3.1 Shape and Size

A nematode has a simple, constant body plan. The body is:

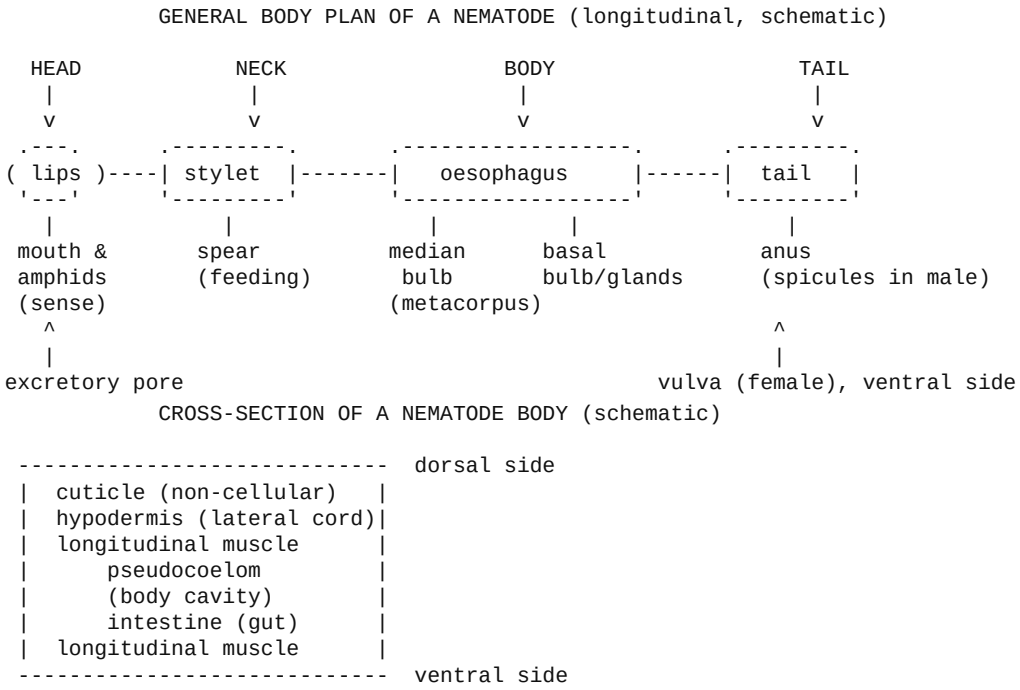
- Vermiform (worm-like), cylindrical and round in cross-section.
- Unsegmented and bilaterally symmetrical.
- Tapering towards both ends - the head (anterior) and the tail (posterior).
- Covered by a tough, non-living cuticle.

Plant-parasitic nematodes are small. Most fall in the range 0.2-3 mm in length; a few ectoparasites such as Longidorus may reach about 5-8 mm. The largest nematode known, Placentonema gigantissima from the sperm whale, is about 8 m long, but no plant parasite comes close to this.

3.2 Body Regions

The body is divided into four regions, and it is useful to learn them because most descriptions in taxonomy refer to them.

- Head or lip region - carries the mouth, lips and the amphids (chemosensory organs).
- Neck or oesophageal region - contains the oesophagus (pharynx) and the nerve ring.
- Body or mid-region - contains the intestine, the reproductive organs, and the excretory and nervous systems.
- Tail - the part behind the anus; carries the phasmids (in Secernentea) and, in males, the copulatory structures.



3.3 The Body Wall

The body wall has three layers: the cuticle (outermost), the hypodermis, and the somatic (body-wall) muscles.

3.3.1 Cuticle

The cuticle is a non-cellular, tough, flexible covering secreted by the hypodermis. It is made mainly of collagen-like proteins and is built of several layers - epicuticle, exocuticle, mesocuticle (median layer), endocuticle and one or more fibre layers.

Functions of the cuticle:

- Protects the body and gives it shape.
- Acts as a permeability barrier.
- Provides a firm point of anchorage for the body muscles, so that the animal can move.
- Bears sense organs and, in plant parasites, the feeding stylet.

Cuticular markings and structures:

- Transverse striae (annulations) - ring-like markings across the body; their number and spacing are used in identification.
- Lateral field - a smooth band on each side, marked by two to four longitudinal lines called incisures.
- Stylet (spear) - a hollow, protrusible feeding organ in Tylenchida; an odontostyle in Dorylaimida.
- Amphids - paired chemosensory organs on the head.
- Phasmids - paired glandular sense organs near the tail in the class Secernentea.
- Deirids, setae and papillae - small sense structures.

3.3.2 Hypodermis

The hypodermis is a thin, living layer just below the cuticle. It secretes the cuticle, and it thickens into four longitudinal cords - two lateral, one dorsal and one ventral. The lateral cords often carry the lateral field. The dorsal and ventral cords carry nerve cords.

3.3.3 Somatic (body-wall) muscles

The body-wall muscles lie beneath the hypodermis. They are all longitudinal and are arranged in four quadrants, separated by the four hypodermal cords. Because only longitudinal muscles exist, a nematode can only bend dorsoventrally; it cannot crawl like an earthworm. Each muscle cell has a contractile part (spindle) and a non-contractile part (sarcoplasm).

3.4 Pseudocoelom

Between the body wall and the gut lies the pseudocoelom, a fluid-filled body cavity. It is not lined by a true mesodermal epithelium (hence "pseudo"). The pseudocoelomic fluid bathes the internal organs and works as a hydrostatic skeleton, giving the body its internal pressure and helping in movement and in support of the gut and reproductive organs.

3.5 Sexual Dimorphism

In most nematodes the sexes are separate, and males and females look different. This difference is called sexual dimorphism.

- Females are usually longer, thicker and straighter; in death the body remains almost straight or gently curved.

- Males are usually smaller, and the tail is curved ventrally (a "walking-stick" shape). Males carry the copulatory organs - a pair of spicules, a gubernaculum, and often a bursa.
- In sedentary parasites the difference is extreme: the female of *Meloidogyne* becomes pear-shaped and the female of *Heterodera* becomes lemon-shaped, while the male stays slender and worm-like.

Feature	Female	Male
Size	Usually longer and thicker	Usually smaller
Body shape	Straight or gently curved	Tail curved ventrally
Tail end	Simple, pointed or rounded	Curved, bears spicules and gubernaculum
Reproductive opening	Vulva on ventral side	Cloaca (common opening of gut and reproductive tract)
Special structures	Ovary, uterus, spermatheca	Spicules, gubernaculum, bursa
Sedentary forms	Swollen (pear-, lemon- or kidney-shaped)	Slender, vermiform

3.6 Measurements Used in Taxonomy (de Man's Indices)

The size and proportions of a nematode are described by a set of standard ratios, called de Man's indices. They are used in every species description.

Index	Meaning
L	Total body length
W	Greatest body width
a	Body length divided by greatest body width (L / W)
b	Body length divided by oesophageal length (L / b')
c	Body length divided by tail length (L / c')
c'	Tail length divided by anal (cloacal) body width
V	Position of the vulva from the anterior end, as a percentage of body length
T	Length of the testis, as a percentage of body length

Worked example: computing de Man's indices

A female root-knot nematode (measured under the microscope) has body length L = 1.00 mm, greatest body width W = 0.040 mm, oesophageal length = 0.150 mm and tail length = 0.050 mm.

1. a = L / W = 1.00 / 0.040 = **25**
2. b = L / oesophageal length = 1.00 / 0.150 = **6.7**
3. c = L / tail length = 1.00 / 0.050 = **20**

So the specimen would be described as "L = 1.00 mm; a = 25; b = 6.7; c = 20". If the anal body width were 0.025 mm, then c' = 0.050 / 0.025 = **2.0**. These values are then compared with published values to confirm the species.

> *In an examination, always show the formula first, then the substitution, then the answer with its unit. The indices a, b and c are pure ratios and carry no unit.*

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3.7 Cuticular Ornamentation and Its Taxonomic Value

The surface of the cuticle is not plain. Its markings are used to identify nematodes:

- Annules (transverse striae) - ring-like markings; their number and shape are characteristic of a species.
- Lateral field - a smooth band on each side, marked by two to four longitudinal incisures; the number of incisures is a useful character.
- Punctations and ridges - small pits or longitudinal ridges on the cuticle of some genera, such as the ring nematodes.
- Cephalic framework - a hardened ring of cuticle around the head that supports the stylet.

3.8 Variation in Body Form

Although the basic body plan is constant, body form varies a great deal between groups:

- Free-living and most plant-parasitic nematodes are vermiform (worm-like) throughout life.
- In sedentary females the body swells after infection: pear-shaped in *Meloidogyne*, lemon-shaped in *Heterodera*, kidney-shaped in *Rotylenchulus*, and club-shaped in *Tylenchulus*.
- The male usually remains vermiform even when the female swells.
- Some species form a "dauer" (survival) juvenile that is resistant to drying.

Exam focus - Lecture 3

Key points. (1) A nematode is vermiform, cylindrical, unsegmented, bilaterally symmetrical and tapers at both ends. (2) The four body regions are head (lip region), neck (oesophageal region), body and tail. (3) The body wall has three layers: cuticle, hypodermis and longitudinal somatic muscles. (4) The cuticle is non-cellular, layered and secreted by the hypodermis; it protects the body and anchors the muscles. (5) Only longitudinal muscles are present, so the animal bends dorsoventrally only. (6) The pseudocoelom is a fluid-filled cavity that acts as a hydrostatic skeleton. (7) Sexual dimorphism is clear: males are smaller, have a curved tail and carry spicules, a gubernaculum and a bursa. (8) de Man's indices (L, W, a, b, c, c', V, T) describe size and proportions.

Short-answer questions.

Q1. Name the three layers of the nematode body wall. *Model answer:* Cuticle (outermost, non-cellular), hypodermis (secretes the cuticle and forms the cords) and somatic longitudinal muscles (four quadrants).

Q2. Why can a nematode only bend dorsoventrally? *Model answer:* Because the body-wall muscles are all longitudinal and arranged in four quadrants; there are no circular muscles, so the body can bend but cannot perform peristaltic crawling.

Q3. What is the function of the pseudocoelom? *Model answer:* It is a fluid-filled cavity that bathes the organs, acts as a hydrostatic skeleton, and supports the gut and reproductive organs during movement.

Long-answer questions.

Q1. Describe the general morphology of a nematode with the help of a labelled diagram. *Outline:*

shape and size - four body regions - body wall (cuticle, hypodermis, muscles) - cuticular structures (stylet, amphids, phasmids, lateral field) - pseudocoelom - digestive and reproductive organs in outline - sexual dimorphism - diagram.

Q2. Explain sexual dimorphism in nematodes with examples. *Outline:* definition - female features (larger, straight, vulva) - male features (smaller, curved tail, spicules, gubernaculum, bursa) - extreme dimorphism in sedentary forms (Meloidogyne pear-shaped, Heterodera lemon-shaped) - taxonomic importance.

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Lecture 4: Anatomy - Digestive, Excretory and Nervous Systems and Sense Organs

4.1 The Digestive System

The digestive system of a nematode is a straight, complete tube running from the mouth at the anterior end to the anus at the posterior end. It has four main parts: the stoma (mouth and buccal cavity), the oesophagus (pharynx), the intestine, and the rectum with the anus.

4.1.1 Mouth and feeding apparatus

The mouth is surrounded by lips and opens into the buccal cavity. In plant-parasitic nematodes the buccal cavity is modified into a hollow feeding spear:

- Stylet (stomatostylet) - found in the order Tylenchida (class Secernentea). It is a hollow spear, often with basal knobs, that can be pushed forward out of the mouth to pierce a plant cell. Saliva (containing enzymes) is injected through it, and the liquefied cell contents are sucked back.
- Odontostyle - found in the order Dorylaimida (class Adenophorea), for example in *Xiphinema* and *Longidorus*. It is a hollow spear formed from a modified tooth.

The stylet is the key organ of plant parasitism. Its length and shape (with or without knobs) are important in identification and also decide how deep the nematode can feed inside the root.

4.1.2 The oesophagus (pharynx)

The oesophagus is a muscular tube that pumps food into the intestine. In the order Tylenchida it is built of four parts:

- Procorpus - the narrow front part.
- Metacorpus (median bulb) - a swollen, muscular bulb with a valve; it pumps the food.
- Isthmus - a narrow connecting part.
- Basal bulb - a glandular bulb that holds the oesophageal glands.

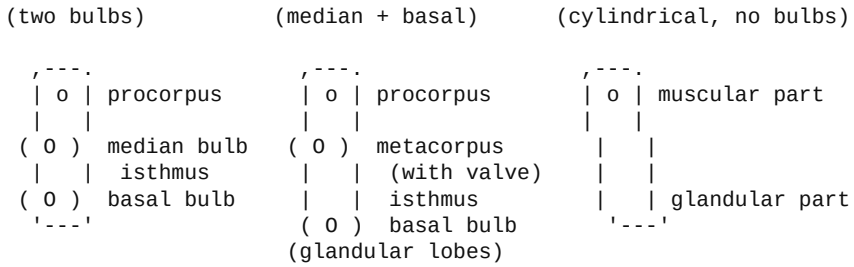
The oesophageal glands are usually three in number - one dorsal and two subventral. They secrete enzymes (such as cellulase and pectinase) that are injected into the plant cell through the stylet to digest cell contents.

Oesophageal types (used in classification):

- Tylenchid (tylenchoid) - median bulb with a valve, followed by a basal bulb; typical of *Tylenchus* and the Tylenchida.
- Rhabditiform - two bulbs (a median bulb and a terminal bulb); typical of the Rhabditida.
- Aphelenchoid - a median bulb followed by a glandular lobe rather than a distinct basal bulb; seen in *Aphelenchoides*.
- Hoplolaimid - the basal bulb has strongly overlapping glandular lobes; seen in *Hoplolaimus*, *Rotylenchulus*.
- Dorylaimoid - a cylindrical oesophagus, muscular at the front and glandular behind, with no bulbs; seen in *Xiphinema* and *Longidorus*.

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        TYPES OF OESOPHAGUS (schematic)
A. RHABDITIFORM       B. TYLENCHID          C. DORYLAIMOID
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Part of the gut	Function
Mouth and buccal cavity (with stylet)	Piercing the plant cell and taking in food
Oesophagus (pharynx)	Sucking and pumping food; site of the oesophageal glands that secrete enzymes
Intestine	Digestion is completed and nutrients are absorbed through a single layer of cells
Rectum and anus	Temporary storage and expulsion of undigested waste

4.1.3 The intestine

The intestine is a simple tube made of a single layer of epithelial cells. Its cells have finger-like projections (microvilli) that increase the surface for absorption. Food is absorbed here, and the undigested residue passes back to the rectum. In some nematodes the intestine has a blind pouch (caecal diverticulum).

4.1.4 Rectum and anus

The hindgut ends in a short rectum that opens to the outside through the anus. In the male the rectum and the reproductive duct join to form a common chamber, the cloaca, through which both faeces and sperm pass.

4.2 The Excretory System

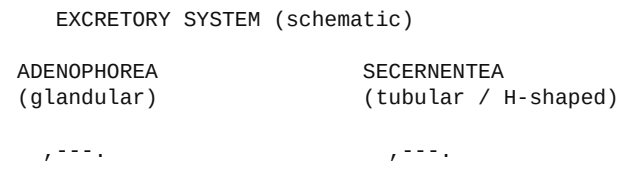
The excretory system removes nitrogenous waste and regulates the water and salt balance (osmoregulation) of the body. Its structure is one of the features that separates the two classes of nematodes.

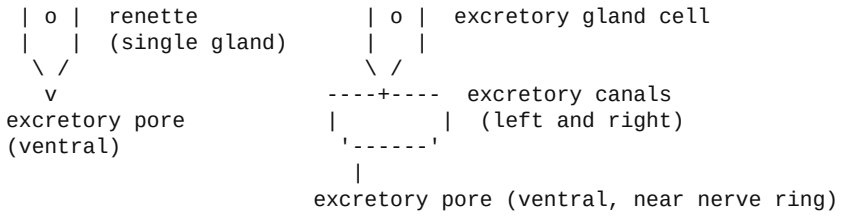
4.2.1 In Adenophorea (Aphasmidia)

The excretory organ is a single, large gland cell called the renette. It lies in the pseudocoelom near the oesophagus and connects to the outside by a short duct and a ventral excretory pore. In some species the renette is absent.

4.2.2 In Secernentea (Phasmidia)

The excretory system is tubular and glandular. There are one or two excretory gland cells, from which canals run forwards and backwards along the sides of the body. The canals join a common duct that opens through the excretory pore on the ventral side, usually near the nerve ring. The whole structure often looks like an "H" and is therefore called the H-shaped excretory system.





4.3 The Nervous System

4.3.1 Central nervous system

The central nervous system is built around a nerve ring that encircles the oesophagus, usually at the level of the median bulb. The nerve ring is the main coordinating centre and is often called the "brain" of the nematode. From the nerve ring, nerves run forwards to the head sense organs and backwards along the body in the form of longitudinal nerve cords:

- Dorsal nerve cord - in the dorsal hypodermal cord.
- Ventral nerve cord - in the ventral hypodermal cord; the largest and most important.
- Lateral nerve cords - in the lateral hypodermal cords.

The cords carry ganglia (groups of nerve cells) and run the length of the body, giving off branches to the muscles, gut, and sense organs.

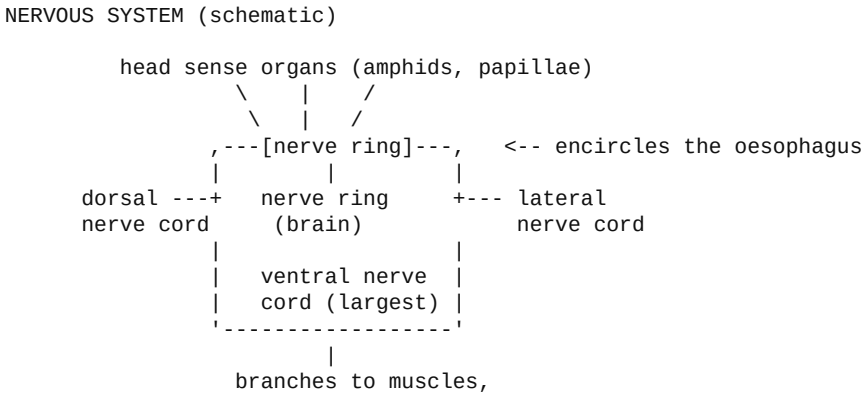
4.3.2 Sense organs

Nematodes have no eyes in the usual sense (except a few free-living aquatic forms that have simple pigment spots called ocelli). Their sense organs are mainly chemical and mechanical:

Location	Function
On the head, on the lips	Chemoreception (smell and taste); shape is used in clas
Near the tail (only in Secernentea)	Chemoreception and glandular secretion
On the head and along the body	Touch (mechanoreception) and chemical sense
On the lateral field	Sensory (mechanoreception)
In a few free-living aquatic forms	Photoreception (light sense)

4.3.3 Importance of the amphids

The amphids are the most important sense organs. They detect chemicals released by plant roots (root exudates) and so help the nematode find a host. Their shape - spiral, circular, pocket-like or slit-like - is a major character in nematode taxonomy, especially in the Adenophorea.



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                    gut and sense organs
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4.4 Osmoregulation and the Role of the Excretory System

The pseudocoelomic fluid is under pressure, and the nematode must keep its water and salt balance steady, especially when the soil dries or floods. The excretory system carries out this osmoregulation as well as the removal of nitrogenous waste. When a nematode is placed in water, the excretory system works actively to expel the excess water that enters; when the soil dries, the nematode reduces its activity and enters a resting state.

4.5 Behaviour and Host Finding

A nematode finds its host mainly by chemical sensing. Root exudates diffuse through the soil water, and the amphids detect the chemical gradient. The nematode then moves up the gradient towards the root. Movement is powered by the longitudinal muscles working against the hydrostatic pressure of the pseudocoelom, giving the characteristic wriggling motion. Once at the root, the stylet and the secretions of the oesophageal glands allow the nematode to pierce a cell and feed.

Exam focus - Lecture 4

Key points. (1) The digestive tract is a complete tube: mouth/stoma, oesophagus, intestine, rectum and anus. (2) Plant parasites feed through a hollow stylet (Tylenchida) or odontostyle (Dorylaimida). (3) In the Tylenchida the oesophagus has a procorpus, a valvate median bulb (metacorpus), an isthmus and a basal bulb, and it bears three oesophageal glands. (4) The intestine has one layer of cells with microvilli for absorption. (5) Adenophorea have a glandular excretory system (renette); Secernentea have a tubular, H-shaped system. (6) The nervous system centres on a nerve ring around the oesophagus, with dorsal, ventral and lateral nerve cords. (7) The main sense organs are the amphids (chemoreception) and the phasmids (only in Secernentea), plus papillae, setae and deirids.

Short-answer questions.

Q1. What is a stylet and what is its function? *Model answer:* A hollow, protrusible feeding spear in the buccal cavity of Tylenchida. It pierces the plant cell wall, saliva with enzymes is injected through it, and the liquefied contents are sucked back.

Q2. Distinguish between the excretory systems of Adenophorea and Secernentea. *Model answer:* Adenophorea have a glandular renette (one gland cell) opening through a ventral pore; Secernentea have a tubular, H-shaped system of canals and gland cells.

Q3. Name the main sense organs of a nematode. *Model answer:* Amphids (head, chemoreception), phasmids (tail, only in Secernentea), papillae and setae (touch), deirids, and ocelli in a few free-living forms.

Long-answer questions.

Q1. Describe the digestive system of a plant-parasitic nematode and the types of oesophagus. *Outline:* parts of the tract - mouth and stylet types - procorpus, median bulb, isthmus, basal bulb - oesophageal glands and enzymes - intestine and rectum - oesophageal types (tylenchid, rhabditiform, aphelenchoid, hoplolaimid, dorylaimoid) - diagram.

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Q2. Explain the nervous system and sense organs of nematodes. *Outline:* nerve ring around the oesophagus - nerve cords (dorsal, ventral, lateral) and ganglia - peripheral branches - sense organs: amphids, phasmids, papillae, setae, deirids, ocelli - importance of amphids in host finding and taxonomy - diagram.

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Lecture 5: Anatomy - Reproductive System

5.1 General Features of Reproduction

Most nematodes are dioecious - that is, the sexes are separate, and each individual is either a male or a female. Reproduction is usually sexual (cross-fertilisation), but other modes also occur:

- Amphimixis - normal cross-fertilisation between male and female.
- Parthenogenesis - development of an egg without fertilisation; common in *Meloidogyne* (mitotic parthenogenesis in several species).
- Hermaphroditism - one individual produces both eggs and sperm; seen in *Caenorhabditis elegans*.

The reproductive organs are tubular and lie in the pseudocoelom, filling much of the body cavity in the adult.

5.2 The Male Reproductive System

5.2.1 The testis and ducts

The male has one testis (monorchic) in most plant-parasitic species, or two testes (diorchic) in some. The testis is usually telogonic, meaning that new germ cells are produced at the tip and the older, more mature stages lie nearer the outlet. The sperm pass along a vas deferens and a seminal vesicle (where sperm are stored) and then through an ejaculatory duct into the cloaca.

5.2.2 Accessory copulatory structures

- Spicules - a pair of cuticular, needle-like structures used to transfer sperm into the female. They are the most reliable sex character.
- Gubernaculum - a cuticular thickening that guides the spicules during copulation.
- Bursa (copulatory bursa) - a flap of cuticle in some males that holds the female during mating; it may reach almost to the tail tip (peloderan) or cover only part of the tail (leptoderan).
- Supplementary organs and caudal alae - additional cuticular structures on the male tail, useful in identification.

Nematode sperm have no flagellum. They move by an amoeboid (crawling) motion.

5.3 The Female Reproductive System

5.3.1 The ovary and ducts

The female reproductive system is a tube that may be single or paired. Its parts, from the free end to the opening, are: ovary, oviduct, uterus, spermatheca (sperm store), vagina and vulva.

- Monodelphic - a single ovary and uterus (for example, in *Tylenchulus* and *Rotylenchulus*).
- Didelphic - two ovaries and two uteri (for example, in *Meloidogyne*, *Heterodera*, *Pratylenchus*).

The relative position of the ovaries is described by special terms: prodelphic (ovary directed forwards), opisthodelphic (directed backwards) and amphidelphic (one of each). Ovaries are

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often reflexed (bent back on themselves) to save space.

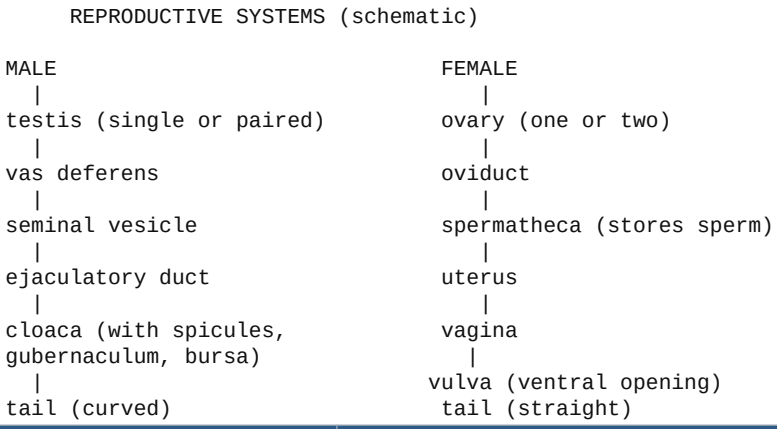
5.3.2 Vulva and egg laying

The vulva opens on the ventral side; its position is given by the index V (distance from the anterior end as a percentage of body length). Eggs may be:

- Laid singly in the soil (for example, many ectoparasites).
- Laid in a gelatinous matrix or egg mass outside the body (*Meloidogyne*; the egg mass is attached to the root surface or pushed into the root).
- Retained inside the dead female body, which hardens to form a cyst (*Heterodera* and *Globodera*).

5.3.3 The egg

A nematode egg is oval and has a shell of several layers - an outer vitelline membrane, a middle chitinous layer and an inner lipid layer. The shell protects the embryo against drying and chemicals, which is one reason why cyst and root-knot nematodes are so hard to kill.



Term	Meaning
Monorchic / diorchic	Male with one / two testes
Monodelphic / didelphic	Female with one / two ovaries
Prodelphic / opisthodelphic	Ovary directed forwards / backwards
Amphidelphic	One ovary directed forwards and one backwards
Telogonic	Germ cells produced at the tip of the gonad
Parthenogenesis	Development of an egg without fertilisation
Gelatinous matrix (egg mass)	Jelly-like mass in which Meloidogyne lays eggs
Cyst	Hardened dead female body of Heterodera/Globodera that holds eggs

5.4 Modes of Reproduction and Sex Determination

- Sexual reproduction (amphimixis) is the rule in most genera, such as *Pratylenchus*, *Radopholus*, *Xiphinema* and *Heterodera*.
- Parthenogenesis is common in *Meloidogyne*; several important species reproduce without males, so a single juvenile can start an infestation.
- Hermaphroditism occurs in some Rhabditida, including the model organism *C. elegans*.

- Sex determination may be genetic or environmental. In *Meloidogyne*, factors such as crowding, food supply and host resistance can shift the sex ratio; when conditions are poor, more males are produced.

5.5 Development and the Life Cycle

Nematode development follows a fixed pattern. After the egg is laid, the embryo develops into the first-stage juvenile (J1) inside the egg. There are four juvenile stages (J1, J2, J3, J4), separated by moults (ecdysis), and then the adult. Each moult produces a new cuticle and the old one is shed.

1. Egg - laid in soil or in an egg mass/cyst.
2. J1 - develops inside the egg; the first moult usually occurs inside the egg.
3. J2 - the infective stage; it hatches and searches for a host.
4. J3 and J4 - feeding and growth stages; the nematode increases in size and the reproductive organs develop.
5. Adult male and female - after the final moult; they mate and the female lays eggs, completing the cycle.

The J2 is the stage that infects plants in the root-knot and cyst nematodes. The time from egg to egg (generation time) is about 3-6 weeks in root-knot nematodes under warm conditions, so several generations can build up in a single crop season.

5.6 Reproductive Features of Important Genera

Genus	Reproductive features
Meloidogyne	Didelphic; parthenogenetic in most species; eggs laid in a gelatinous matrix (egg mass)
Heterodera	Didelphic; sexual; eggs retained in the body, which becomes a cyst
Globodera	Didelphic; sexual; females become round cysts
Tylenchulus	Monodelphic; sexual; eggs in a gelatinous matrix on the root surface
Rotylenchulus	Monodelphic; males rare, largely parthenogenetic; eggs in a gelatinous matrix
Radopholus	Didelphic; sexual and parthenogenetic; eggs laid singly in the root tissue
Pratylenchus	Didelphic; sexual; eggs laid singly in the root and soil

5.7 Population Growth and the Reproduction Factor

When a nematode feeds on a good host, the population increases many times in one crop season. The increase is measured by the reproduction factor, Rf = final population (Pf) divided by initial population (Pi). If Rf is greater than 1, the crop is a good host and the population has increased; if Rf is less than 1, the crop is a poor host and the population has fallen. This idea is used to test varieties for resistance and to plan crop rotations.

Exam focus - Lecture 5

Key points. (1) Nematodes are mostly dioecious and reproduce sexually; parthenogenesis is common in Meloidogyne and hermaphroditism in C. elegans. (2) The male has a testis (monorchic or diorchic), vas deferens, seminal vesicle and cloaca, with spicules, gubernaculum and often a bursa. (3) The female has ovary, oviduct, uterus, spermatheca, vagina and vulva; it may be monodelphic or

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didelphic. (4) Eggs have a multi-layered shell and may be laid singly, in an egg mass (Meloidogyne) or retained in a cyst (Heterodera). (5) Development passes through four juvenile stages and four moults; the J2 is the infective stage. (6) Generation time in root-knot nematodes is about 3-6 weeks.

Short-answer questions.

Q1. What are spicules and the gubernaculum? *Model answer:* Spicules are a pair of cuticular, needle-like structures in the male used to transfer sperm; the gubernaculum is a cuticular guide that steers the spicules during copulation.

Q2. Distinguish monodelphic from didelphic females. *Model answer:* A monodelphic female has a single ovary and uterus (e.g., Tylenchulus, Rotylenchulus); a didelphic female has two (e.g., Meloidogyne, Heterodera, Pratylenchus).

Q3. Name the infective juvenile stage in root-knot and cyst nematodes. *Model answer:* The second-stage juvenile (J2); it hatches from the egg and infects the root.

Long-answer questions.

Q1. Describe the female reproductive system of a nematode and the different ways eggs are laid. *Outline:* ovary, oviduct, uterus, spermatheca, vagina, vulva - monodelphic vs didelphic - egg shell layers - eggs laid singly, in an egg mass (Meloidogyne) or retained in a cyst (Heterodera) - diagram.

Q2. Explain the male reproductive system and the life cycle of a nematode. *Outline:* testis (monorchic/diorchic), ducts, cloaca - accessory structures: spicules, gubernaculum, bursa - sperm without flagella - life cycle: egg, J1-J4 with moults, adult - J2 as infective stage - generation time - diagram.

Lecture 6: Biology of Economically Important Nematode Genera - Meloidogyne, Heterodera, Tylenchulus, Rotylenchulus and Radopholus

6.1 Introduction

These five genera cause most of the economic damage from plant-parasitic nematodes in India and worldwide. They are grouped by the way they feed:

- Sedentary endoparasites (the nematode enters the root, becomes fixed in one place and feeds from specialised cells): *Meloidogyne* and *Heterodera*.
- Semi-endoparasites (part of the body stays in the root and part remains outside): *Tylenchulus* and *Rotylenchulus*.
- Migratory endoparasite (the nematode moves through the root tissues as it feeds): *Radopholus*.

6.2 Meloidogyne - Root-Knot Nematode

6.2.1 Position and species

Meloidogyne belongs to the family Heteroderidae (order Tylenchida). About 100 species are known. The most important are *M. incognita*, *M. javanica*, *M. arenaria* and *M. hapla*. In India, *M. incognita* and *M. javanica* are the commonest on vegetables and pulses.

6.2.2 Biology and life cycle

The infective second-stage juvenile (J2) enters the root just behind the root tip and moves between the cells until it reaches the vascular cylinder. There it becomes fixed and induces the formation of special feeding cells called giant cells (nurse cells), which supply it with food. The surrounding root tissue swells into a gall or root knot.

```
            LIFE CYCLE OF MELOIDOGYNE (root-knot nematode)

Egg mass (gelatinous matrix) on the root surface
     |
     v
Egg  -->  J1 (develops inside egg)  -->  J2 hatches out (INFECTIVE)
                                                |
                                                v
                J2 enters the root just behind the root tip
                                                |
                                                v
                J2 moves between cells to the vascular cylinder
                                                |
                                                v
                Giant cells (nurse cells) + gall (root knot) are formed
                                                |
                                                v
                J2 -> J3 -> J4  (feeding and swelling)
                                                |
                                                v
                Adult female (pear-shaped, sedentary)   /   Adult male (vermiform, leaves root)
                                                |
                                                v
                Female lays eggs in a gelatinous matrix  --> Egg mass
                (parthenogenetic in many species)
```

6.2.3 Symptoms and host range

Symptoms: knobby galls (root knots) on the roots, forked and stubby roots, poor root system, stunted and yellow plants, and wilting even when the soil is moist. Above ground the damage looks like a general nutrient-deficiency problem, which is why it is often missed.

Host range: very wide - tomato, brinjal, okra, chilli, cucurbits, potato, pulses (blackgram, greengram, cowpea), groundnut, cotton, tobacco, banana, tea, coffee and many ornamentals. In Tamil Nadu, root-knot is a major problem in tomato, brinjal and bhendi.

6.3 Heterodera - Cyst Nematode

6.3.1 Position and species

Heterodera also belongs to the family Heteroderidae and is a sedentary endoparasite. Important species are *H. avenae* (cereal cyst nematode, the cause of "molya" disease of wheat in Rajasthan), *H. schachtii* (sugar-beet cyst), *H. glycines* (soybean cyst), *H. cajani* (pigeonpea cyst) and *H. oryzicola* (rice cyst).

6.3.2 Biology

The J2 enters the root and becomes fixed near the vascular cylinder, where it induces multinucleate feeding cells called syncytia (transfer cells). The female swells into a lemon shape, breaks out through the root surface, and remains attached by her head. When she dies, her body wall tans and hardens into a brown cyst, which contains 200-600 eggs. The cyst is the main survival structure and can protect eggs in soil for several years.

6.3.3 Symptoms and host range

Symptoms: stunted plants in patches, poor tillering in cereals, and small white-to-brown lemon-shaped cysts visible on the roots (and later free in the soil). Infestation is usually patchy in the field.

Host range: cereals (wheat, barley, rice, maize), sugar beet, soybean, pigeonpea, potato (the related genus *Globodera*), and several crucifers.

6.4 Tylenchulus - Citrus Nematode

6.4.1 Position and biology

Tylenchulus semipenetrans (family Tylenchulidae) is a semi-endoparasite. Only the female infects. Her slender anterior end is inserted into the root cortex, where she feeds on a small group of nurse cells, while the swollen posterior part remains outside the root. She lays eggs in a gelatinous matrix at the root surface. The female is monodelphic.

6.4.2 Symptoms and host range

Symptoms: "citrus slow decline" - the tree slowly loses vigour, leaves become sparse and yellow, twigs die back, and fruit size and yield fall over several years. The decline is gradual, hence the name.

Host range: mainly citrus (all citrus species and rootstocks), and also grape, olive and persimmon. It is the most important nematode of citrus in India.

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6.5 Rotylenchulus - Reniform Nematode

6.5.1 Position and biology

Rotylenchulus reniformis (family Hoplolaimidae) is a semi-endoparasite. The young female enters the root cortex with her anterior end while the posterior part stays outside, and she then swells into a kidney or reniform (bean-shaped) form. She is monodelphic and lays eggs in a gelatinous matrix. Males are rare and reproduction is largely parthenogenetic.

6.5.2 Symptoms and host range

Symptoms: stunting, poor growth, yellowish foliage and reduced yield; the symptoms are not very specific and resemble nutrient deficiency.

Host range: very wide - cotton, pulses (pigeonpea, blackgram, greengram), vegetables (tomato, brinjal, okra), oilseeds (castor, groundnut) and many weeds. In India it is important on cotton and pulses.

6.6 Radopholus - Burrowing Nematode

6.6.1 Position and biology

Radopholus similis (family Pratylenchidae) is a migratory endoparasite. Both males and females enter the root and tunnel through the cortex, feeding on cells and leaving cavities behind. Because it moves and feeds continuously, it destroys large areas of root tissue. The male has a bursa and reproduction may be sexual or by parthenogenesis.

6.6.2 Symptoms and host range

Symptoms: dark lesions and cavities in the roots, root death, and in banana the classic "toppling disease" (also called blackhead toppling), in which the plant falls over because the roots can no longer anchor it. In citrus it causes "spreading decline".

Host range: banana, plantain, citrus, black pepper, coconut, tea and several ornamentals. In India it is a major constraint in banana and black pepper.

6.7 Comparison of the Five Genera

Genus	Feeding type	Key feature	Main symptom	Important hosts
Meloidogyne	Sedentary endoparasite	Pear-shaped female; giant cells; egg mass	Root galls (knots), stunting, wilting	Tomato, brinjal, okra, pulses, groundnut, banana
Heterodera	Sedentary endoparasite	Lemon-shaped female that becomes a cyst; syncytia	Stunting in patches, cysts on roots	Wheat, barley, rice, soybean, pigeonpea
Tylenchulus	Semi-endoparasite	Anterior in root, posterior outside; monodelphic	Citrus slow decline	Citrus, grape, olive
Rotylenchulus	Semi-endoparasite	Kidney (reniform) female; monodelphic	General stunting, poor growth	Cotton, pulses, vegetables, oilseeds
Radopholus	Migratory endoparasite	Tunnels through root cortex; cavities	Root lesions, banana toppling	Banana, citrus, black pepper, coconut

The feeding type is a good memory aid: Meloidogyne and Heterodera enter and settle (endoparasites), Tylenchulus and Rotylenchulus enter only halfway (semi-endoparasites), and Radopholus keeps moving

(migratory endoparasite).

6.8 Survival and Dissemination

- Meloidogyne survives as eggs in egg masses and in soil debris; the eggs resist drying and many chemicals.
- Heterodera and Globodera survive as cysts, which protect the eggs for several years in the soil.
- Tylenchulus and Rotylenchulus survive as eggs in the soil and in the gelatinous matrix.
- Radopholus survives in living roots, in planting material (banana suckers, corms) and in soil.

All these nematodes are spread by infested soil, water, farm implements, and above all by planting material such as suckers, seedlings and nursery stock.

6.9 The Host-Parasite Relationship

The relationship between these nematodes and their host is more than simple feeding. The sedentary endoparasites (*Meloidogyne*, *Heterodera*) change the plant itself: they induce giant cells or syncytia, which are living cells with many nuclei and dense cytoplasm, that act as nutrient sinks. The plant's nutrients are diverted to these cells, and the plant's growth hormones are disturbed, producing galls and stunting. The semi-endoparasites (*Tylenchulus*, *Rotylenchulus*) feed from a small group of cells at the point of entry, and the migratory endoparasite *Radopholus* simply destroys the cells it passes through. This difference explains why root-knot and cyst nematodes cause galls and stunting, while the burrowing nematode causes root rot and toppling.

Exam focus - Lecture 6

Key points. (1) Meloidogyne is a sedentary endoparasite; the J2 induces giant cells and galls; the female is pear-shaped and lays eggs in a gelatinous matrix; it is parthenogenetic. (2) Heterodera is a sedentary endoparasite; it induces syncytia and the female becomes a lemon-shaped cyst holding 200-600 eggs. (3) Tylenchulus semipenetrans is a semi-endoparasite causing citrus slow decline; the female is monodelphic. (4) Rotylenchulus reniformis is a semi-endoparasite with a reniform (kidney-shaped) female and a very wide host range. (5) Radopholus similis is a migratory endoparasite causing root cavities and banana toppling disease. (6) Galls are typical of root-knot; cysts are typical of Heterodera; slow decline is typical of citrus.

Short-answer questions.

Q1. Why is Meloidogyne called a sedentary endoparasite? *Model answer:* The infective J2 enters the root, moves to the vascular cylinder and then becomes permanently fixed (sedentary) while feeding from giant cells; the swollen female never leaves the root.

Q2. What is a cyst and which genus forms it? *Model answer:* A cyst is the hardened, tanned body wall of a dead Heterodera (or Globodera) female, which holds 200-600 eggs and protects them in the soil for years.

Q3. Name the nematode that causes toppling disease of banana. *Model answer:* Radopholus similis, the burrowing nematode; it tunnels through the root cortex and destroys the root system.

Long-answer questions.

Q1. Describe the biology, symptoms and host range of Meloidogyne with a labelled life-cycle diagram. *Outline:* position and species - J2 infection and migration - giant cells and gall formation - swollen pear-shaped female and egg mass - parthenogenesis - life cycle 3-6 weeks - symptoms (galls, stunting, wilting) - host range - Tamil Nadu examples - diagram.

Q2. Compare the five economically important nematode genera. *Outline:* feeding types (sedentary endoparasite, semi-endoparasite, migratory endoparasite) - for each genus: female shape, special feeding cells, key symptom, important hosts - present as a comparison table - management note.

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Lecture 7: Taxonomy of Plant-Parasitic Nematodes of the Secernentea and Adenophorea

7.1 What Taxonomy Means

Taxonomy is the science of identifying, naming and classifying organisms. It has three parts:

- Identification - finding out which species a specimen belongs to.
- Classification - arranging species into groups based on shared characters.
- Nomenclature - giving each species a scientific name under the International Code of Zoological Nomenclature (genus + species, both in italics).

7.2 Basis of Classification

Nematodes are classified mainly on morphology, supported now by molecular data. The characters used include:

- Presence or absence of phasmids.
- Shape and size of the amphids.
- Type of oesophagus and arrangement of oesophageal glands.
- Type of excretory system (glandular or tubular).
- Cuticle features (annulations, lateral field).
- Presence or absence of caudal glands in the tail.
- Type of stylet (stomatostylet or odontostyle).
- Molecular characters - DNA sequences (18S and 28S rRNA genes, and mitochondrial DNA).

7.3 The Two Classical Classes

In 1933 Chitwood and Chitwood divided the phylum Nematoda into two classes, based chiefly on the presence or absence of phasmids:

- Class Secernentea (also called Phasmidia) - phasmids present.
- Class Adenophorea (also called Aphasmidia) - phasmids absent.

This simple split is still used as the first step in identifying a plant-parasitic nematode.

Character	Secernentea (Phasmidia)	Adenophorea (Aphasmidia)
Phasmids	Present (near the tail)	Absent
Amphids	Small, pore-like or circular	Large, pocket-like, spiral or slit-like
Excretory system	Tubular, H-shaped, with canals	Glandular, with a renette; may be absent
Caudal glands	Absent	Usually present
Cuticle	Usually annulated; lateral field often present	Often smooth; different ornamentation
Oesophageal glands	Few (often 3), opening into the oesophagus	More numerous, opening into the oesophagus
Main habitat	Mostly terrestrial and freshwater	Mostly marine and freshwater; some soil

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Character	Secernentea (Phasmidia)	Adenophorea (Aphasmidia)
Plant-parasitic orders	Tylenchida (the great majority)	Dorylaimida, Triplonchida (Trichodorida)
Example genera	Meloidogyne, Heterodera, Pratylenchus, Radopholus, Tylenchulus, Rotylenchulus	Xiphinema, Longidorus, Trichodorus, Paratrichodorus

7.4 Orders and Families of Plant-Parasitic Nematodes

7.4.1 Class Secernentea

Almost all plant-parasitic nematodes of the Secernentea belong to the order Tylenchida. The order is divided into two suborders - Tylenchina and Aphelenchina - and then into families.

Family	Important genera and note
Tylenchidae	Tylenchus; mainly root-hair and fungal feeders, weakly parasitic
Anguinidae	Anguina (seed-gall nematode), Ditylenchus (stem and bulb nematode)
Pratylenchidae	Pratylenchus (lesion), Radopholus (burrowing)
Hoplolaimidae	Hoplolaimus, Helicotylenchus, Rotylenchulus (reniform), Scutellonema
Heteroderidae	Heterodera (cyst), Globodera (potato cyst), Meloidogyne (root-knot)
Tylenchulidae	Tylenchulus (citrus nematode), Paratylenchus (pin nematode)
Belonolaimidae	Belonolaimus (sting nematode)
Criconematidae	Criconemoides, Hemicycliophora (ring and sheath nematodes)

7.4.2 Class Adenophorea

Two orders contain plant-parasitic nematodes, and they are important as ectoparasites and as virus vectors.

Family	Important genera and note
Longidoridae (order Dorylaimida)	Xiphinema (dagger nematode), Longidorus (needle nematode); long odontostyle; vectors of Nepoviruses
Trichodoridae (order Triplonchida / Trichodorida)	Trichodorus, Paratrichodorus (stubby-root nematodes); short curved odontostyle; vectors of Tobraviruses

7.5 The Modern Classification

The classical two-class system has been refined by molecular studies, especially by De Ley and Blaxter (2002, 2004). In the modern scheme:

- The phylum Nematoda is divided into the classes Chromadorea and Enoplea.
- The old Secernentea (including the whole of the Tylenchida) falls mainly in Chromadorea, and the Adenophorea falls in Enoplea.
- The phylum is further split into a series of clades (Clades 1-12), based on gene sequences.
- In the modern scheme the plant-parasitic Tylenchida are placed within an expanded order Rhabditida (as the infraorder Tylenchomorpha).

For the syllabus and for examination answers, the classical Secernentea and Adenophorea split is the one expected, with the modern view given as a note.

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                TAXONOMIC HIERARCHY (example: root-knot nematode)

Phylum      :  Nematoda
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Class       :  Secernentea (Chromadorea in modern scheme)
Order       :  Tylenchida (Rhabditida - Tylenchomorpha, modern)
Suborder    :  Tylenchina
Superfamily :  Tylenchoidea / Hoplolaimoidea
Family      :  Heteroderidae
Genus       :  Meloidogyne
Species     :  Meloidogyne incognita
```

7.6 Diagnostic Characters Used in Keys

When identifying a plant-parasitic nematode, the following characters are examined in order:

- The stylet - its presence, length, shape and whether it has knobs.
- The oesophagus - the type (tylenchid, dorylaimoid and so on) and the arrangement of the glands.
- The amphids - their shape and position.
- The phasmids - present or absent, and their position.
- The female reproductive system - monodelphic or didelphic, and the position of the ovary.
- The tail shape and the male characters - spicules, gubernaculum and bursa.
- Morphometric ratios (de Man's indices) and, increasingly, DNA sequences.

7.7 Nomenclature and Type Specimens

Scientific names follow the International Code of Zoological Nomenclature. A new species must be described in a recognised journal and a type specimen deposited in a museum or collection. The genus name takes a capital letter and the species name a small letter, and both are italicised (for example, *Meloidogyne incognita*). The author's name and year are added at the first mention.

Exam focus - Lecture 7

Key points. (1) Taxonomy is identification, classification and nomenclature. (2) The classical split (Chitwood and Chitwood, 1933) is based mainly on phasmids: Secernentea (Phasmidia) have phasmids; Adenophorea (Aphasmidia) do not. (3) Secernentea have small amphids, a tubular (H-shaped) excretory system, and no caudal glands; Adenophorea have large amphids, a glandular renette, and caudal glands. (4) Almost all plant-parasitic nematodes of the Secernentea are in the order Tylenchida (families Tylenchidae, Anguinidae, Pratylenchidae, Hoplolaimidae, Heteroderidae, Tylenchulidae). (5) Plant-parasitic Adenophorea belong to the Dorylaimida (Xiphinema, Longidorus) and Triplonchida (Trichodorus). (6) The modern scheme uses the classes Chromadorea and Enoplea and a series of clades.

Short-answer questions.

Q1. On what character did Chitwood and Chitwood base their two classes? *Model answer:* On the presence or absence of phasmids: Secernentea (Phasmidia) have phasmids; Adenophorea (Aphasmidia) do not.

Q2. Give two differences between Secernentea and Adenophorea. *Model answer:* Secernentea have small pore-like amphids and a tubular H-shaped excretory system; Adenophorea have large pocket-like or spiral amphids and a glandular renette.

Q3. Name two virus-vector nematode genera and their families. *Model answer:* Xiphinema and

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Longidorus (family Longidoridae, order Dorylaimida), which transmit Nepoviruses; and Trichodorus and Paratrichodorus (family Trichodoridae), which transmit Tobraviruses.

Long-answer questions.

Q1. Describe the classification of plant-parasitic nematodes into Secernentea and Adenophorea. *Outline:* basis of classification (phasmids, amphids, excretory system, caudal glands) - characters of Secernentea - characters of Adenophorea - comparison table - orders and families of each - example genera.

Q2. Give an account of the families and genera of plant-parasitic nematodes. *Outline:* order Tylenchida and its families with representative genera - order Dorylaimida (Longidoridae) - order Triplonchida (Trichodoridae) - modern classification into Chromadorea and Enoplea - taxonomic hierarchy of Meloidogyne as an example.

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Lecture 8: Classification of Plant-Parasitic Nematodes Based on Feeding Habits; Beneficial Nematodes

8.1 Why Classify by Feeding Habit

The feeding habit of a nematode decides where it lives, how it damages the plant, and therefore how it must be managed. Classifying nematodes by feeding habit is thus both scientific and practical. Two questions fix the group: (1) does the nematode stay outside the root or enter it, and (2) if it enters, does it keep moving or settle down?

8.2 Feeding-Habit Groups of Plant-Parasitic Nematodes

Group	Meaning	Position	Examples
Ectoparasites	Stay outside the root and feed on the surface cells	Outside the root	Xiphinema, Longidorus, Trichodorus, Belonolaimus, Criconemoides
Migratory endoparasites	Enter the root, tunnel through the tissue and keep moving	Inside, mobile	Pratylenchus, Radopholus
Sedentary endoparasites	Enter the root, become fixed and induce feeding cells	Inside, fixed	Meloidogyne, Heterodera, Globodera, Nacobbus
Semi-endoparasites	Anterior end inside the root, posterior outside	Partly inside	Tylenchulus, Rotylenchulus, Hoplolaimus
Above-ground (foliar) feeders	Feed on stem, leaf, flower or seed	Aerial parts	Anguina, Ditylenchus, Aphelenchoides, Bursaphelenchus

The root-knot and cyst nematodes are the classic sedentary endoparasites, because they induce giant cells or syncytia and then stop moving. Radopholus and Pratylenchus are the classic migratory endoparasites, because they keep moving and leave a trail of damaged cells.

8.3 Other Ways of Grouping Nematodes

8.3.1 By the part of the plant attacked

- Root feeders - most species (*Meloidogyne*, *Heterodera*, *Pratylenchus* and others).
- Stem and bulb feeders - *Ditylenchus*.
- Foliar feeders - *Aphelenchoides* (leaf and bud), *Anguina* (seed gall).
- Vector nematodes - *Xiphinema*, *Longidorus*, *Trichodorus*, which also transmit plant viruses.

8.3.2 By feeding type in the soil food web (free-living forms)

- Bacterivores - feed on bacteria.
- Fungivores - feed on fungi.
- Predators - attack other nematodes and small soil animals.
- Omnivores - feed on a mixture of food.
- Insect parasites - feed on insects and are beneficial (see 8.4).

8.4 Beneficial Nematodes

Not all nematodes are pests. Some are useful to agriculture, and a few are of great value in science.

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8.4.1 Entomopathogenic nematodes (insect-killing nematodes)

These are the best-known beneficial nematodes. They carry symbiotic bacteria and kill insect pests.

- *Steinernema* species carry the bacterium *Xenorhabdus*.
- *Heterorhabditis* species carry the bacterium *Photorhabdus*.

The infective juvenile enters the insect through natural openings, releases the bacterium into the insect's blood, and the insect dies within one or two days. The nematodes then reproduce inside the dead insect and emerge to search for new hosts. Formulations are sold as biopesticides against borers, cutworms, white grubs and other soil insects. *Heterorhabditis indica* is an Indian isolate widely studied and used.

8.4.2 Nematodes used against other pests

- *Deladenus* (*Beddingia*) *siricidicola* sterilises the female *Sirex* wood wasp and has been used to control this forestry pest.
- *Romanomermis* species parasitise mosquito larvae and have been tested for mosquito control.
- *Phasmarhabditis hermaphrodita* is used commercially against slugs.

8.4.3 Nematodes useful in science and soil health

- *Caenorhabditis elegans* is a model organism in genetics, development and neurobiology (Nobel Prize, 2002).
- Free-living nematodes are used as bioindicators of soil health, because the mix of feeding groups changes with farming practice and pollution.
- Free-living nematodes also help decompose organic matter and release plant nutrients, so they support soil fertility.

Beneficial nematode	Use / role
Steinernema, Heterorhabditis	Entomopathogenic; biopesticides against insect pests (with symbiotic bacteria)
Heterorhabditis indica	Indian isolate used as a biopesticide against soil insects
Deladenus (Beddingia) siricidicola	Sterilises the Sirex wood wasp (forestry pest control)
Romanomermis	Parasitises mosquito larvae (mosquito control)
Phasmarhabditis hermaphrodita	Biological control of slugs
Caenorhabditis elegans	Model organism in biological research
Free-living soil nematodes	Bioindicators of soil health; nutrient cycling

8.5 Nematodes and Soil Health

Because free-living nematodes occupy every level of the soil food web, the nematode community is a sensitive indicator of soil health. Two simple measures are used:

- The maturity index - based on the proportion of coloniser (opportunist) and persister (sensitive) nematodes; a high value indicates a stable, undisturbed soil.
- The ratio of fungivores to bacterivores, and the proportion of predatory nematodes, which indicate the condition of the food web.

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A soil rich in organic matter and undisturbed usually has a diverse nematode community with many persister and predatory forms. Heavy tillage, chemical use and low organic matter shift the community towards opportunists. This is why nematodes are called "indicators of soil health".

Exam focus - Lecture 8

Key points. (1) Feeding habit is decided by whether the nematode enters the root and whether it keeps moving. (2) The five groups are ectoparasites, migratory endoparasites, sedentary endoparasites, semi-endoparasites and above-ground feeders. (3) Ectoparasites: Xiphinema, Longidorus, Trichodorus; migratory endoparasites: Pratylenchus, Radopholus; sedentary endoparasites: Meloidogyne, Heterodera; semi-endoparasites: Tylenchulus, Rotylenchulus; above-ground: Anguina, Ditylenchus, Aphelenchoides. (4) Beneficial nematodes include the entomopathogenic Steinernema and Heterorhabditis (biopesticides), Deladenus, Romanomermis, and the model organism C. elegans. (5) Free-living nematodes are bioindicators of soil health.

Short-answer questions.

Q1. Give the five feeding-habit groups of plant-parasitic nematodes with one example each. *Model answer:* Ectoparasite - Xiphinema; migratory endoparasite - Pratylenchus; sedentary endoparasite - Meloidogyne; semi-endoparasite - Rotylenchulus; above-ground feeder - Ditylenchus.

Q2. How do entomopathogenic nematodes kill insects? *Model answer:* The infective juvenile carries symbiotic bacteria (Xenorhabdus in Steinernema, Photorhabdus in Heterorhabditis); it enters the insect and releases the bacterium, which kills the insect within one or two days.

Q3. Give two uses of beneficial nematodes. *Model answer:* As biopesticides against insect pests (Steinernema, Heterorhabditis) and as bioindicators of soil health (free-living nematodes); C. elegans is a model organism.

Long-answer questions.

Q1. Classify plant-parasitic nematodes according to feeding habits and give examples of each group. *Outline:* basis of classification - ectoparasites - migratory endoparasites - sedentary endoparasites - semi-endoparasites - above-ground feeders - comparison table - practical importance of the grouping.

Q2. Write an account of beneficial nematodes. *Outline:* entomopathogenic nematodes and their symbiotic bacteria - Steinernema and Heterorhabditis as biopesticides - Deladenus against Sirex - Romanomermis against mosquitoes - C. elegans as a model organism - free-living nematodes as bioindicators and in nutrient cycling.

Lecture 9: Mid Semester Examination

9.1 Scope of the Examination

No new theory is prescribed for this lecture. The Mid Semester Examination covers Lectures 1 to 8. This chapter is a revision aid: it lists the points most often asked and gives a quick-recall table. Use it to check that nothing has been left out before the examination.

9.2 Revision Checklist (Lectures 1-8)

- Lecture 1 - Nematology: definition; Needham (1743) and *Anguina tritici*; *Heterodera schachtii* (1859-71); *Meloidogyne* (1889, genus 1949); N. A. Cobb (1913), father of nematology; Chitwood and Chitwood (1933); development in India (IARI, late 1950s; root-knot on tea, 1901).
- Lecture 2 - Position in the animal kingdom; characters of the phylum; two classes; importance to humans, animals, plants and insects; habitat and diversity; global loss US $80-125 billion (10-14 per cent).
- Lecture 3 - General morphology; four body regions; body wall (cuticle, hypodermis, muscles); pseudocoelom; sexual dimorphism; de Man's indices.
- Lecture 4 - Digestive system and oesophageal types; excretory system (renette vs tubular); nervous system (nerve ring, nerve cords); sense organs (amphids, phasmids).
- Lecture 5 - Male and female reproductive systems; spicules, gubernaculum, bursa; monodelphic vs didelphic; egg shell; life cycle and moults; J2 as infective stage.
- Lecture 6 - Biology of *Meloidogyne*, *Heterodera*, *Tylenchulus*, *Rotylenchulus* and *Radopholus*; giant cells, syncytia, cysts, egg mass; the comparison table.
- Lecture 7 - Secernentea vs Adenophorea; basis of classification; orders and families; modern Chromadorea/Enoplea scheme.
- Lecture 8 - Feeding-habit groups; beneficial nematodes (Steinernema, Heterorhabditis, C. elegans).

9.3 Quick-Recall Table

Question	One-line answer
Father of nematology	N. A. Cobb (1913)
First plant-parasitic nematode recorded	Anguina tritici (Needham, 1743, wheat)
Two classes of Nematoda	Secernentea (Phasmidia) and Adenophorea (Aphasmidia)
Basis of the two-class split	Presence or absence of phasmids
Infective stage in root-knot/cyst nematodes	Second-stage juvenile (J2)
Feeding cells of Meloidogyne	Giant cells (nurse cells)
Feeding cells of Heterodera	Syncytia (transfer cells)
Survival structure of Heterodera	Cyst (200-600 eggs)
Nematode causing citrus slow decline	Tylenchulus semipenetrans

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Question	One-line answer
Nematode causing banana toppling	Radopholus similis
Nematode causing molya of wheat	Heterodera avenae
Virus-vector nematodes	Xiphinema, Longidorus (Nepoviruses); Trichodorus (Tobraviruses)
Entomopathogenic nematodes	Steinernema, Heterorhabditis
Model organism	Caenorhabditis elegans
de Man's index for body length / width	a

9.4 Suggested Examination Pattern

The Mid Semester Examination is usually a written paper of about one and a half to two hours, with a mix of very short answer, short answer and long answer questions, and a choice within each section. Answers should follow the sequence: definition, then points, then a labelled diagram or table wherever one is asked for.

9.5 Common Mistakes in the Examination

- Writing the history without years and names - always give Needham (1743), Cobb (1913), Chitwood and Chitwood (1933).
- Confusing the two classes - remember Secernentea has phasmids, Adenophorea does not.
- Mixing up feeding cells - giant cells belong to Meloidogyne, syncytia to Heterodera.
- Forgetting to label diagrams - an unlabelled diagram loses marks.
- Writing long paragraphs where points are asked - use short, numbered points.
- Ignoring numerical questions - show the formula, the substitution and the answer with its unit.

Exam focus - Lecture 9

Key points. (1) The examination covers Lectures 1-8. (2) Learn the historical milestones with their years and names. (3) Learn the two-class classification and its basis. (4) Learn the body regions, body wall and de Man's indices. (5) Learn the digestive, excretory and nervous systems and the sense organs. (6) Learn the male and female reproductive systems and the life cycle. (7) Learn the biology and symptoms of the five key genera. (8) Learn the feeding-habit groups and the beneficial nematodes.

Short-answer questions.

Q1. Write the systematic position of plant-parasitic nematodes in two lines. *Model answer:* Kingdom Animalia, phylum Nematoda; class Secernentea (order Tylenchida) for most plant parasites, and class Adenophorea (orders Dorylaimida and Triplonchida) for the ectoparasitic virus vectors.

Q2. Define de Man's index a. *Model answer:* a is the body length divided by the greatest body width (L / W); it is a pure ratio with no unit.

Q3. What is the infective stage of Meloidogyne and where does it enter the root? *Model answer:* The second-stage juvenile (J2), which enters the root just behind the root tip and migrates to the vascular cylinder.

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Long-answer questions.

Q1. Write short notes on any four of the following: (a) N. A. Cobb; (b) giant cells; (c) the two classes of nematodes; (d) egg mass; (e) beneficial nematodes. *Outline:* for each - definition, one or two key facts, one example or diagram.

Q2. Describe the general anatomy of a nematode, covering the body wall and the digestive, excretory, nervous and reproductive systems. *Outline:* body regions - body wall - digestive system and oesophageal types - excretory system - nervous system and sense organs - male and female reproductive systems - labelled diagrams.

Lecture 10: Symptoms of Nematode Damage and Role of Nematodes in Disease Development

10.1 Why Nematode Symptoms Are Often Missed

Plant-parasitic nematodes live mostly in the soil and attack the roots, so the early damage is invisible. The above-ground symptoms they cause - stunting, yellowing and wilting - look exactly like nutrient deficiency, drought or poor soil. For this reason nematodes are called the "hidden enemy". Two clues usually point to nematodes:

- The damage appears in patches (a "nematode patch") rather than evenly across the field, because nematode populations are patchy.
- The damage is worse on light, sandy soils and in fields cropped repeatedly with a susceptible crop.

10.2 Above-Ground Symptoms

- Stunting - plants are shorter and smaller than normal.
- Chlorosis - leaves turn pale green to yellow.
- Wilting - plants wilt even when the soil has enough moisture (because the root system is damaged).
- Poor tillering and fewer branches in cereals and pulses.
- Reduced leaf size, sparse foliage and dieback in perennial crops ("slow decline").
- Reduced yield and poor quality of produce.
- Nutrient-deficiency-like appearance - the plant looks starved although fertiliser has been applied.

10.3 Below-Ground Symptoms

- Galls or root knots - swellings on the roots (*Meloidogyne*).
- Cysts - small lemon-shaped bodies on the roots, later free in the soil (*Heterodera*, *Globodera*).
- Lesions - dark, sunken areas on the roots (*Pratylenchus*).
- Root necrosis and decay, often with secondary rotting organisms.
- Stubby or forked roots, and necrosis of root tips (*Belonolaimus*, ectoparasites).
- Cavities and tunnels inside the cortex (*Radopholus*).
- Excessive branching of roots (proliferation), sometimes with shortened roots.
- Complete loss of the anchoring root system, leading to toppling (banana).

10.4 Symptoms of Important Nematode Groups

Nematode group	Characteristic symptom	Example
Root-knot nematodes	Galls (knots) on roots; stunting and wilting	Meloidogyne on tomato
Cyst nematodes	Lemon-shaped cysts on roots; patchy stunting	Heterodera on wheat ("molya")

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Nematode group	Characteristic symptom	Example
Lesion nematodes	Dark lesions and decay on roots	Pratylenchus on many crops
Burrowing nematode	Cavities in roots; toppling of the plant	Radopholus on banana
Ectoparasites	Stubby roots, necrosis of root tips	Trichodorus, Belonolaimus
Foliar nematodes	Discoloured, distorted leaves; "white tip" of rice	Aphelenchoides besseyi on rice
Stem and bulb nematodes	Swelling, distortion and rot of stem or bulb	Ditylenchus dipsaci on onion
Seed-gall nematode	Galls in place of grain (ear cockle)	Anguina tritici on wheat
Virus-vector nematodes	Virus-like symptoms (mottling, fanleaf)	Xiphinema and grapevine fanleaf

10.5 Damage Symptoms by Crop Group

Crop group	Typical nematode damage symptoms
Cereals (rice, wheat, maize)	Patchy stunting, poor tillering, galls on upland rice roots, "molya" in wheat, white tip of rice, ear cockle in wheat
Vegetables (tomato, brinjal, okra, chilli)	Root galls, stunting, yellowing, wilting in the afternoon, reduced fruit set
Pulses (pigeonpea, chickpea, blackgram)	Stunting, poor podding, root galls, cysts on roots, wilt complexes
Oilseeds (groundnut, sunflower)	Stunted plants, galls and lesions on roots, poor pod filling
Fibre crops (cotton, jute)	Patchy stunting, reduced boll or fibre yield
Fruit and plantation crops (banana, citrus, coconut)	Slow decline, dieback, toppling (banana), sparse foliage, reduced fruit size

10.6 Role of Nematodes in Disease Development

Nematodes damage plants in two ways - directly and indirectly.

10.6.1 Direct damage

- Mechanical damage - the stylet punctures cells and migratory nematodes tunnel through tissues.
- Physiological damage - sedentary nematodes induce giant cells or syncytia, which divert nutrients and disturb the plant's growth hormones; this produces galls and stunting.
- Secretion of enzymes and other substances - cellulases and pectinases break down cell walls; other secretions change the plant's metabolism.

10.6.2 Indirect damage - the nematode as a predisposing agent

This is the more important role in many field problems. Nematodes seldom act alone. They:

- Make wounds that let fungi and bacteria enter the root.
- Change root exudates and plant physiology, making the plant more susceptible to other pathogens.
- Break down host resistance to a fungus or bacterium, so a resistant variety becomes diseased.

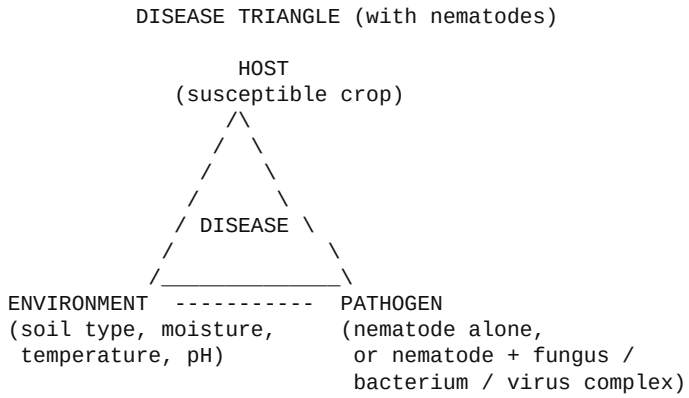
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- Act as vectors, carrying viruses from plant to plant.

The result is a disease complex, in which nematode + fungus (or bacterium, or virus) together cause far more damage than either alone. This is discussed in detail in Lecture 11.

10.6.3 The disease triangle with nematodes

Disease develops only when a susceptible host, a virulent pathogen and a favourable environment come together. For nematodes the "pathogen" side of the triangle often contains a complex rather than a single organism.



10.8 Confirming a Nematode Problem

Field symptoms only suggest a nematode problem; they do not prove it. Confirmation needs:

- A soil sample (composite, from the root zone) for extraction and counting of nematodes.
- A root sample to look for galls, cysts, lesions or cavities.
- A bioassay - grow a susceptible plant in the suspect soil and check its roots.
- Comparison with a healthy part of the field, to separate nematode damage from nutrient or water problems.

Once the nematode is identified and counted, the population is compared with the damage threshold to decide whether management is needed.

Exam focus - Lecture 10

Key points. (1) Nematodes are the "hidden enemy" because the damage is below ground and the above-ground symptoms mimic nutrient deficiency. (2) Above-ground symptoms: stunting, chlorosis, wilting, patchy poor growth, reduced yield. (3) Below-ground symptoms: galls, cysts, lesions, necrosis, stubby or forked roots, cavities, toppling. (4) Damage appears in patches because nematode populations are patchy. (5) Nematodes cause direct damage (mechanical, physiological, enzymatic) and indirect damage by predisposing the plant to other pathogens. (6) The disease triangle needs a susceptible host, a virulent pathogen and a favourable environment; the pathogen side is often a nematode + fungus/bacterium/virus complex.

Short-answer questions.

Q1. Why is nematode damage usually patchy in a field? *Model answer:* Because nematode populations are themselves patchy in the soil (they spread slowly and survive in localised pockets), so the damage shows up as patches rather than evenly.

Q2. Give four below-ground symptoms of nematode damage. *Model answer:* Root galls/knots, cysts

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on roots, dark lesions and necrosis, and stubby or forked roots (also cavities and toppling).

Q3. What is meant by a nematode as a "predisposing agent"? *Model answer:* A nematode that does not itself kill the plant but makes it more susceptible to a fungus, bacterium or virus, usually by wounding the root and changing its physiology.

Long-answer questions.

Q1. Describe the symptoms of nematode damage in crop plants. *Outline:* why symptoms are missed (hidden enemy, patchiness) - above-ground symptoms - below-ground symptoms - symptoms by nematode group - symptoms by crop group - table and examples.

Q2. Explain the role of nematodes in disease development. *Outline:* direct damage (mechanical, physiological, enzymatic) - indirect damage and disease complexes - nematodes as vectors - the disease triangle with nematodes - one or two examples (root-knot + Fusarium wilt; root-knot + Ralstonia) - diagram.

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Lecture 11: Interaction Between Plant-Parasitic Nematodes and Disease-Causing Fungi, Bacteria and Viruses

11.1 The Concept of a Disease Complex

In the field, nematodes rarely act alone. They commonly occur together with fungi, bacteria or viruses, and the combined effect on the crop can be far greater than the sum of the two pathogens acting separately. This combined effect is called a disease complex or an interaction.

Interactions are of three kinds:

- Additive - the total damage is roughly the sum of the damage caused by each pathogen alone.
- Synergistic - the total damage is much greater than the sum; the nematode makes the disease worse. This is the most important type in agriculture.
- Antagonistic - one pathogen reduces the effect of the other.

11.2 Nematode-Fungus Interactions

This is the best known and most damaging type of interaction. The classic example is the wilt complex, in which a root-knot nematode and a wilt fungus act together.

- *Meloidogyne* + *Fusarium oxysporum* - causes severe wilt in tomato, cotton, chickpea and pigeonpea. Plants of a variety that is resistant to *Fusarium* alone may wilt when the root-knot nematode is also present, because the nematode breaks down resistance.
- *Meloidogyne* + *Rhizoctonia solani* - root-rot complex in many vegetables.
- *Meloidogyne* + *Pythium* or *Phytophthora* - damping-off and root-rot complex in nurseries.
- *Heterodera* + *Fusarium* - poor growth and wilt in cereals and other crops.
- *Pratylenchus* + *Verticillium* - early dying disease of potato.
- *Radopholus* + *Fusarium* - blackhead and root rot of banana.

How the nematode helps the fungus:

- It makes wounds through which the fungus enters.
- It changes root exudates and cell chemistry, which encourages fungal growth.
- It causes physiological stress, so the plant's defences are weakened.

11.3 Nematode-Bacterium Interactions

- *Meloidogyne* + *Ralstonia solanacearum* (bacterial wilt) - a serious complex in tomato and brinjal. The nematode galls provide entry points and the wilt develops earlier and more severely.
- *Meloidogyne* + *Agrobacterium tumefaciens* (crown gall) - the nematode damage increases gall formation.
- *Meloidogyne* + *Clavibacter* (*Corynebacterium*) - bacterial canker and wilt complexes.
- *Pratylenchus* + *Ralstonia* - wilt complex in potato and other crops.

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The mechanism is the same as for fungi: the nematode wounds the root and alters the plant's physiology, so the bacterium establishes more easily and the disease is worse.

11.4 Nematode-Virus Interactions

Some nematodes are vectors - they carry plant viruses from one plant to another while feeding. The virus is picked up on the feeding spear and transmitted to the next plant.

Nematode genus	Virus group transmitted	Important example
Xiphinema (dagger)	Nepoviruses	Grapevine fanleaf virus; tomato ringspot virus
Longidorus (needle)	Nepoviruses	Raspberry ringspot virus; tomato black ring virus
Trichodorus / Paratrichodorus (stubby-root)	Tobraviruses	Tobacco rattle virus; pea early browning virus

How transmission works:

- While feeding, the nematode takes up virus particles along with plant sap; the virus sticks to the lining of the odontostyle.
- The nematode moves to a healthy plant and, while feeding, releases the virus into the plant.
- The virus may be retained for weeks to months; in some vectors it is retained through moulting, in others it is lost at moult.

Grapevine fanleaf, spread by Xiphinema index, is one of the most damaging virus diseases of grapevine, and it is a good Indian and world example of nematode-virus interaction.

11.5 Summary of Interactions

Interaction	Example	Effect
Nematode + fungus	Meloidogyne + Fusarium oxysporum	Severe wilt in tomato, cotton, chickpea
Nematode + fungus	Pratylenchus + Verticillium	Early dying of potato
Nematode + fungus	Meloidogyne + Rhizoctonia/Pythium	Root rot and damping-off complexes
Nematode + bacterium	Meloidogyne + Ralstonia solanacearum	Bacterial wilt of tomato and brinjal
Nematode + bacterium	Meloidogyne + Agrobacterium	Increased crown gall
Nematode + virus	Xiphinema + Nepovirus	Grapevine fanleaf
Nematode + virus	Trichodorus + Tobravirus	Tobacco rattle disease

11.6 Importance of Interactions in Management

Interactions change the way a crop must be managed:

- A variety that is resistant to a fungus may still fail if the nematode is present, so resistance must be checked against the whole complex.
- Controlling the nematode often reduces the fungus or bacterium disease as well, so nematode management becomes part of disease management.
- Crop rotation must consider both the nematode and the fungal or bacterial pathogen, because a rotation that removes one may allow the other.

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11.7 Case Studies of Disease Complexes in India

- **Wilt of tomato.** Root-knot nematode (*Meloidogyne incognita*) with *Fusarium oxysporum* f. sp. *lycopersici*, and sometimes with *Ralstonia solanacearum*. Plants wilt and die even in resistant varieties when the nematode is present.
- **Wilt of cotton.** *Meloidogyne incognita* or *Rotylenchulus reniformis* with *Fusarium oxysporum* f. sp. *vasinfectum*; the nematode increases the wilt and reduces the stand.
- **Wilt of chickpea and pigeonpea.** *Meloidogyne* or *Heterodera cajani* with *Fusarium* wilt; a major cause of patchy crop failure in central India.
- **Bacterial wilt of brinjal and tomato.** *Meloidogyne* with *Ralstonia solanacearum*; the disease appears earlier and spreads faster where root-knot is present.
- **Potato early dying.** *Pratylenchus* with *Verticillium*; the crop dies down early and yield falls.

In all these cases, controlling the nematode is an important part of controlling the disease. This is why nematode management and disease management must be planned together.

Exam focus - Lecture 11

Key points. (1) A disease complex is the combined effect of two or more pathogens; interactions may be additive, synergistic or antagonistic. (2) Synergistic interaction (nematode + fungus/bacterium) is the most damaging. (3) Classic fungal complex: Meloidogyne + Fusarium oxysporum causing wilt in tomato, cotton and chickpea. (4) Classic bacterial complex: Meloidogyne + Ralstonia solanacearum causing bacterial wilt of tomato and brinjal. (5) Nematode-virus interaction works through vector transmission: Xiphinema and Longidorus (Nepoviruses), Trichodorus and Paratrichodorus (Tobraviruses). (6) Nematodes act by wounding the root, changing root exudates and breaking down host resistance. (7) Managing the nematode often reduces the associated disease.

Short-answer questions.

Q1. What is a disease complex? *Model answer:* The combined effect on a crop of two or more pathogens (for example a nematode and a fungus) acting together, which may be greater than the sum of their separate effects.

Q2. Give one example each of a nematode-fungus and a nematode-bacterium complex. *Model answer:* Nematode-fungus: Meloidogyne + Fusarium oxysporum causing wilt of tomato. Nematode-bacterium: Meloidogyne + Ralstonia solanacearum causing bacterial wilt of tomato and brinjal.

Q3. Name two nematode vectors and the virus groups they transmit. *Model answer:* Xiphinema (Nepoviruses, e.g., grapevine fanleaf) and Trichodorus (Tobraviruses, e.g., tobacco rattle virus).

Long-answer questions.

Q1. Explain the interactions between plant-parasitic nematodes and fungi, bacteria and viruses. *Outline:* definition of a disease complex and its three types - nematode-fungus interactions with examples - nematode-bacterium interactions with examples - nematode-virus interaction and vector transmission - mechanisms (wounding, root exudates, breaking resistance) - summary table.

Q2. Discuss how nematode interactions affect crop management. *Outline:* synergistic damage -

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breakdown of host resistance - controlling the nematode reduces the associated disease - implications for crop rotation and resistant varieties - examples from tomato, cotton and banana.

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Lecture 12: Nematode Parasites of Cereals

12.1 Introduction

Cereals occupy the largest area of any crop group in India, and several nematodes cause serious losses in them. Rice and wheat are the two most important cereal hosts, followed by maize, sorghum and the millets. Cereal nematode damage is usually patchy and is often mistaken for poor fertility or water stress.

12.2 Nematodes of Rice

12.2.1 Root-knot nematode of rice - Meloidogyne graminicola

This is the most important nematode of rice in India. It is a serious problem in upland (rainfed) and aerobic rice and in nurseries, and is now increasingly seen in irrigated rice as well.

- Symptoms: galls (knots) on the roots, poor root system, stunting, yellowing, and "crop failure in the nursery"; in severe cases plants die in patches.
- It survives in soil as eggs and can attack the next crop.

12.2.2 White tip disease - Aphelenchoides besseyi

The foliar nematode *A. besseyi* causes "white tip" of rice.

- Symptoms: the tips of the leaves turn white or pale, the leaf becomes twisted or distorted, and the panicle is small with poor grain filling.
- It is seed-borne, so infected seed carries the nematode to new fields.

12.2.3 Rice root nematode - Hirschmanniella oryzae and H. mucronata

These are important in lowland (flooded) rice.

- Symptoms: brown necrotic lesions on the roots, reduced root growth, stunting and poor tillering; yield loss can be significant in continuous wetland rice.

12.2.4 Ufra disease - Ditylenchus angustus

This stem and panicle nematode occurs mainly in deepwater and lowland rice in eastern India and Bangladesh.

- Symptoms: distortion of the stem and panicle, empty or partly filled panicles, and a whitish, crinkled appearance at the base of the panicle (ufra).

12.2.5 Rice cyst nematode - Heterodera oryzicola

A cyst nematode of rice reported from parts of India; it causes stunting and cysts on the roots.

12.3 Nematodes of Wheat

12.3.1 Cereal cyst nematode - Heterodera avenae ("molya" disease)

This is the best-known nematode problem of wheat in India, especially in the light soils of Rajasthan, where it is called "molya".

- Symptoms: stunted, pale plants in patches, poor tillering, and small white-to-brown cysts on the roots; severely affected patches may fail completely.
- It also attacks barley and oats; the cysts survive in soil for years.

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12.3.2 Ear cockle or seed-gall nematode - Anguina tritici

This is the historic nematode of wheat (the very first plant-parasitic nematode ever recorded).

- Symptoms: instead of grain, the ear carries dark, hard galls ("cockles"); the galled seed is smaller and lighter.
- The galls are carried with the seed, so clean seed is an important control measure.

12.3.3 Root-lesion nematode - Pratylenchus spp.

Root-lesion nematodes cause brown lesions on wheat roots and reduce vigour, often together with other pathogens.

12.4 Nematodes of Maize, Sorghum and Millets

- *Pratylenchus* spp. - lesion nematodes causing root damage and stunting in maize and sorghum.
- *Meloidogyne* spp. - root-knot nematodes on maize and sorghum.
- *Hoplolaimus* and *Rotylenchulus* - lance and reniform nematodes on sorghum and millets.

Crop	Nematode	Main symptom
Rice	Meloidogyne graminicola	Galls on roots, stunting, nursery failure
Rice	Aphelenchoides besseyi	White tip of leaves, poor grain filling
Rice	Hirschmanniella oryzae	Root necrosis, stunting (wetland rice)
Rice	Ditylenchus angustus	Ufra - distorted stem and panicle
Wheat	Heterodera avenae	Molya - patchy stunting, cysts on roots
Wheat	Anguina tritici	Ear cockle - galls in place of grain
Wheat	Pratylenchus spp.	Root lesions, poor vigour
Maize, sorghum	Pratylenchus, Meloidogyne	Root damage, stunting

Worked example: crop-loss calculation in a cereal field

A wheat field of 1 hectare has a normal (healthy) yield of 40 quintals per hectare. Nematode damage ("molya") affects 30 per cent of the field, where the yield is reduced by 40 per cent; the remaining 70 per cent yields normally.

1. Potential yield of the whole field = 1 ha x 40 q/ha = **40 q**.
2. Yield from the affected area (0.30 ha) = 0.30 x 40 x (1 - 0.40) = 0.30 x 40 x 0.60 = **7.2 q**.
3. Yield from the healthy area (0.70 ha) = 0.70 x 40 = **28 q**.
4. Actual total yield = 7.2 + 28 = **35.2 q**.
5. Total loss = 40 - 35.2 = **4.8 q**, which is 4.8 / 40 x 100 = **12 per cent** of the potential yield.
6. If wheat is valued at Rs 2,000 per quintal (assumed price), the loss = 4.8 x 2,000 = **Rs 9,600** for the hectare.

The price is only an assumption for the calculation; in an examination, state the price you use and keep the steps in the same order.

12.5 Management of Cereal Nematodes

- Use clean, nematode-free seed (very important for ear cockle and white tip).

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- Grow resistant or tolerant varieties where available (for example, against the cereal cyst nematode).
- Rotate with non-host crops - pulses or crucifers in place of continuous wheat break the cyst nematode cycle.
- Summer ploughing and repeated drying of the field reduce nematode populations.
- Flooding the field, where possible, reduces many soil nematodes (but not the wetland-adapted *Hirschmanniella*).
- Apply organic amendments such as neem cake and well-decomposed farmyard manure.
- Use bioagents - *Pseudomonas fluorescens*, *Trichoderma viride* and *Paecilomyces lilacinus* - as seed or soil treatment.
- Apply need-based nematicides (for example, carbofuran 3G at about 1-2 kg a.i./ha) where the population is above the threshold.

12.6 A Simple Rice Nematode Management Calendar

Stage	Action
Before sowing / nursery	Use clean, nematode-free seed; treat seed with Pseudomonas fluorescens and Trichoderma viride
Nursery	Solarise the nursery bed; avoid repeated use of the same wet bed for root-knot
Land preparation	Summer ploughing; incorporate neem cake and organic matter
At transplanting	Use healthy seedlings; avoid seedlings from an infested nursery
Crop growth	Maintain water management; where possible flood, which reduces many soil nematodes
If population is high	Apply a need-based nematicide (carbofuran 3G at about 1-2 kg a.i./ha)
After harvest	Remove and destroy crop debris; sample the soil to plan the next crop

Exam focus - Lecture 12

Key points. (1) The most important nematode of rice is Meloidogyne graminicola (root-knot), which causes galls and nursery failure. (2) Aphelenchoides besseyi causes white tip of rice and is seed-borne. (3) Hirschmanniella oryzae is the rice root nematode of wetland rice. (4) Ditylenchus angustus causes ufra of deepwater rice. (5) Heterodera avenae causes "molya" of wheat, with patchy stunting and cysts on roots. (6) Anguina tritici causes ear cockle of wheat, with galls in place of grain. (7) Management rests on clean seed, resistant varieties, crop rotation, organic amendments, bioagents and need-based nematicides.

Short-answer questions.

Q1. What is molya disease and which nematode causes it? *Model answer:* Molya is a patchy stunting and poor tillering disease of wheat in light soils (notably Rajasthan), caused by the cereal cyst nematode Heterodera avenae.

Q2. Name the nematode that causes white tip of rice and state one management measure. *Model answer:* Aphelenchoides besseyi; it is seed-borne, so use clean, nematode-free seed (hot-water seed treatment also helps).

Q3. Which nematode causes ufra disease of rice? *Model answer:* Ditylenchus angustus, a stem and

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panicle nematode of deepwater and lowland rice.

Long-answer questions.

Q1. Describe the important nematodes of rice and their management. *Outline:* Meloidogyne graminicola (root-knot, galls, nursery failure) - Aphelenchoides besseyi (white tip, seed-borne) - Hirschmanniella oryzae (root necrosis) - Ditylenchus angustus (ufra) - Heterodera oryzicola - management: clean seed, resistant varieties, rotation, flooding, organic matter, bioagents, nematicides.

Q2. Discuss the nematode problems of wheat with suitable control measures. *Outline:* Heterodera avenae and molya - Anguina tritici and ear cockle - Pratylenchus and root lesion - symptoms and losses - management by clean seed, resistant varieties, crop rotation, summer ploughing, bioagents and need-based chemicals - worked loss calculation.

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Lecture 13: Nematode Parasites of Vegetable Crops

13.1 Introduction

Vegetables are grown on small, intensively cropped plots, often year after year, and this favours the build-up of nematodes. Root-knot nematodes (*Meloidogyne* spp.) are the single biggest nematode problem of vegetables in India, followed by the reniform nematode (*Rotylenchulus reniformis*) and lesion nematodes (*Pratylenchus*). Vegetable crops are usually raised in nurseries, and a nematode-infested nursery spreads the problem to the whole field.

13.2 Tomato

Tomato is attacked by several nematodes, and the damage is often worsened by wilt pathogens.

- *Meloidogyne incognita* and *M. javanica* - root galls, stunting, yellowing and wilting; heavy yield loss. Together with *Fusarium oxysporum* they cause a severe wilt complex.
- *Rotylenchulus reniformis* - stunting and poor growth.
- *Pratylenchus* spp. - root lesions.

13.3 Brinjal

- *Meloidogyne* spp. - root galls and wilting; often with *Ralstonia solanacearum* (bacterial wilt), forming a wilt complex.
- *Rotylenchulus reniformis* - stunting.

13.4 Okra (Bhendi)

- *Meloidogyne incognita* - root galls, stunting and poor pod yield.
- *Rotylenchulus reniformis* - stunting and yellowing.

13.5 Chilli

- *Meloidogyne* spp. and *Rotylenchulus reniformis* - stunting, yellowing and reduced fruit yield; also interacts with wilt pathogens.

13.6 Potato

- *Globodera rostochiensis* and *G. pallida* - the golden and pale cyst nematodes, the most damaging nematodes of potato worldwide. The female turns golden-yellow (G. rostochiensis) or cream-white (G. pallida) before becoming a brown cyst. Symptoms: patchy stunting, poor tuber size and yield.
- *Meloidogyne* spp. - root galls and tuber damage.
- *Ditylenchus destructor* - potato tuber rot nematode.

13.7 Onion and Garlic

- *Ditylenchus dipsaci* - the stem and bulb nematode; causes swelling, twisting and rot of the bulb and foliage ("onion bloat"). It survives as a "eelworm wool" in dry bulbs and in soil.
- *Meloidogyne* spp. - root galls.

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13.8 Crucifers (Cabbage, Cauliflower)

- *Meloidogyne* spp. and *Heterodera* spp. - root galls, stunting and poor head formation.

13.9 Cucurbits

- *Meloidogyne* spp. - root galls, stunting and poor fruit set; the first plant-parasitic nematode record (Berkeley, 1855) was actually on cucumber roots.

13.10 Carrot and Other Root Vegetables

- *Meloidogyne* and *Heterodera* - root galls, forking and reduced marketable yield.

Vegetable	Main nematodes	Symptoms
Tomato	Meloidogyne, Rotylenchulus, Pratylenchus	Root galls, stunting, wilt complex
Brinjal	Meloidogyne, Rotylenchulus	Galls, stunting, bacterial wilt complex
Okra	Meloidogyne, Rotylenchulus	Galls, stunting, poor pod yield
Chilli	Meloidogyne, Rotylenchulus	Stunting, yellowing, low fruit yield
Potato	Globodera, Meloidogyne, Ditylenchus	Cysts on roots, tuber damage, poor yield
Onion, garlic	Ditylenchus dipsaci	Bulb swelling and rot (onion bloat)
Crucifers	Meloidogyne, Heterodera	Galls, stunting, poor head formation
Cucurbits	Meloidogyne	Root galls, poor fruit set
Carrot, beet	Meloidogyne, Heterodera	Galls, forked roots, low marketable yield

Worked example: root gall index and threshold decision

A tomato field is sampled to decide whether a nematicide is needed. One hundred plants are lifted at random and the roots are graded on a 0-5 root-gall index:

Gall index	Meaning	Number of plants
0	No galls	30
1	1-2 galls	25
2	3-10 galls	20
3	11-30 galls	15
4	31-100 galls	10
5	More than 100 galls, roots severely galled	0

1. Mean gall index = (0x30 + 1x25 + 2x20 + 3x15 + 4x10 + 5x0) / 100.
2. = (0 + 25 + 40 + 45 + 40 + 0) / 100 = 150 / 100 = **1.5**.
3. The action threshold for root-knot on tomato is usually a mean gall index of about **2.0** (in a 0-5 scale). Since 1.5 is below 2.0, an immediate nematicide application is not justified; the crop should be monitored and the soil sampled again.
4. If the mean gall index had been 2.0 or more (or if the soil count had been above about 2 J2 per gram of soil), the field would be treated - for example with carbofuran 3G at about 1-2 kg a.i./ha, or with a bioagent such as Paecilomyces lilacinus.

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Threshold values vary with crop, cultivar and soil; the figures above are standard published ranges and should be adjusted to local TNAU advisories.

13.11 Management of Vegetable Nematodes

- Raise seedlings in fumigated, solarised or steam-treated nursery beds.
- Use resistant varieties where available; tomato varieties carrying the *Mi* gene resist root-knot nematodes.
- Rotate with non-host crops such as maize, sorghum or cereals.
- Soil solarisation - cover moist soil with transparent polythene (about 25 micrometre thick) for 4-6 weeks in summer.
- Apply organic amendments - neem cake, castor cake and well-decomposed farmyard manure.
- Use bioagents - *Pseudomonas fluorescens*, *Trichoderma viride* and *Paecilomyces lilacinus* as seed, seedling or soil treatment (TNAU recommendation).
- Grow marigold (*Tagetes* spp.) as a trap or intercrop; its root exudates reduce root-knot populations.
- Remove and destroy severely infected plants and crop debris.
- Apply need-based nematicides (carbofuran 3G or phorate 10G at about 1-2 kg a.i./ha) in the soil before planting.

13.12 Nursery Management for Vegetables

Most vegetable nematode problems begin in the nursery, because seedlings are raised on the same beds again and again.

- Solarise or steam the nursery soil before sowing.
- Use fumigated or treated nursery soil, or raise seedlings in trays with sterilised media.
- Treat seed with Pseudomonas fluorescens and Trichoderma viride.
- Avoid overhead watering that keeps the bed constantly wet; good drainage reduces nematode activity.
- Inspect seedlings before transplanting; discard any with galled or poor roots.
- Rotate nursery sites and do not raise the same susceptible crop on the same bed every season.

A clean nursery is the cheapest way to keep root-knot out of a vegetable field. Once galled seedlings are transplanted, the whole field is infested.

Exam focus - Lecture 13

Key points. (1) Root-knot nematodes (Meloidogyne) are the biggest nematode problem of vegetables in India. (2) Tomato, brinjal, okra and chilli suffer root galls, stunting and wilt complexes. (3) Potato is attacked by the golden and pale cyst nematodes (Globodera rostochiensis and G. pallida). (4) Onion and garlic are attacked by Ditylenchus dipsaci (onion bloat). (5) The root-gall index (0-5 scale) is used to judge the severity of root-knot; a mean gall index of about 2.0 is a common action threshold. (6) Management uses resistant varieties (Mi gene in tomato), non-host rotation, solarisation, organic amendments, bioagents (P. fluorescens, T. viride, P. lilacinus) and need-based nematicides.

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Short-answer questions.

Q1. Which nematode causes onion bloat and how does it survive? *Model answer:* Ditylenchus dipsaci, the stem and bulb nematode; it survives as "eelworm wool" (dry, coiled nematodes) in dry bulbs, seed and soil.

Q2. What is the golden cyst nematode and which crop does it attack? *Model answer:* Globodera rostochiensis, which attacks potato; the female turns golden-yellow before hardening into a brown cyst.

Q3. How is a root-gall index used? *Model answer:* Roots are graded 0-5 according to the number of galls; the mean index is compared with an action threshold (about 2.0) to decide whether treatment is needed.

Long-answer questions.

Q1. Describe the nematode problems of important vegetable crops and their management. *Outline:* Meloidogyne on tomato, brinjal, okra, chilli - Rotylenchulus and Pratylenchus - Globodera on potato - Ditylenchus dipsaci on onion and garlic - symptoms - management: resistant varieties, nursery treatment, rotation, solarisation, organic matter, bioagents, marigold, need-based nematicides.

Q2. Explain how the root-gall index and the threshold concept are used in nematode management. *Outline:* the 0-5 gall index - calculation of the mean index - the action threshold (about 2.0) - soil population threshold (about 2 J2/g soil) - decision rule - worked example - choice of control measure.

Lecture 14: Nematode Parasites of Pulses, Oilseed and Fibre Crops

14.1 Introduction

Pulses, oilseeds and fibre crops are grown over large areas in India, often in rotation with cereals. Several nematodes attack them, and the damage is frequently part of a disease complex with wilt fungi and bacteria. The cyst nematodes of pulses, the root-knot nematodes of groundnut and the reniform nematode of cotton are the most important problems in this group.

14.2 Nematodes of Pulses

14.2.1 Pigeonpea (Redgram)

- *Heterodera cajani* - the pigeonpea cyst nematode; causes stunting, poor growth and cysts on the roots; very important in the Deccan and central India.
- *Meloidogyne* spp. - root galls and stunting.
- *Rotylenchulus reniformis* - reniform nematode, causing stunting.
- *Pratylenchus* spp. - root lesions.
- Together with *Fusarium udum*, these nematodes worsen the wilt of pigeonpea.

14.2.2 Chickpea (Bengalgram)

- *Meloidogyne* spp. - root galls.
- *Pratylenchus* spp. and *Rotylenchulus reniformis* - stunting and root damage.
- Root-knot often acts with *Fusarium oxysporum* f. sp. *ciceri* to increase wilt.

14.2.3 Blackgram and Greengram

- *Meloidogyne* spp. - the most important; root galls, stunting and poor podding.
- *Rotylenchulus reniformis* - stunting and yellowing.

14.2.4 Cowpea and Soybean

- *Meloidogyne* spp. on cowpea.
- *Heterodera glycines* - the soybean cyst nematode, a major problem in soybean-growing countries; it also occurs in India.

14.2.5 Lentil and Horsegram

- *Meloidogyne* and *Pratylenchus* cause root damage and poor growth.

14.3 Nematodes of Oilseed Crops

14.3.1 Groundnut

- *Meloidogyne arenaria* and *M. hapla* - root galls and pod damage; a serious problem on light soils.
- *Pratylenchus* spp. - root lesions and pod rot complex.
- *Rotylenchulus reniformis* - stunting.
- *Aphelenchoides* spp. - foliar damage in some situations.

14.3.2 Sunflower

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- *Meloidogyne* spp. and *Rotylenchulus reniformis* - stunting, poor head development.

14.3.3 Mustard and Rapeseed

- *Meloidogyne* spp. and *Heterodera* spp. - root galls, cysts and stunting.

14.3.4 Sesame, Castor and Safflower

- *Meloidogyne* and *Rotylenchulus* cause stunting and poor yield; castor is a good host of *Rotylenchulus reniformis* and should not be used in rotation with cotton for that reason.

14.4 Nematodes of Fibre Crops

14.4.1 Cotton

- *Rotylenchulus reniformis* - the reniform nematode is the most important nematode of cotton in India; it causes stunting, yellowing and reduced boll yield, and it has a very wide host range.
- *Meloidogyne incognita* - root galls and a wilt complex with *Fusarium*.
- *Hoplolaimus* - lance nematode, causing root damage.

14.4.2 Jute

- *Meloidogyne* spp. and *Hoplolaimus* - stunting and poor fibre yield.

14.4.3 Sunn Hemp and Mesta

- *Meloidogyne* spp. - root galls and stunting.

Crop	Main nematodes	Symptoms
Pigeonpea	Heterodera cajani, Meloidogyne, Rotylenchulus	Cysts and galls on roots, stunting, wilt complex
Chickpea	Meloidogyne, Pratylenchus, Rotylenchulus	Galls, root lesions, wilt complex
Blackgram, greengram	Meloidogyne, Rotylenchulus	Galls, stunting, poor podding
Cowpea, soybean	Meloidogyne, Heterodera glycines	Galls, cysts, stunting
Groundnut	Meloidogyne arenaria, Pratylenchus	Galls, root lesions, pod damage
Sunflower	Meloidogyne, Rotylenchulus	Stunting, poor head
Mustard, rapeseed	Meloidogyne, Heterodera	Galls, cysts, stunting
Cotton	Rotylenchulus reniformis, Meloidogyne	Stunting, yellowing, low boll yield
Jute	Meloidogyne, Hoplolaimus	Stunting, low fibre yield

Worked example: planning a crop rotation and calculating the population change

A field is infested with *Meloidogyne incognita*. The initial population is taken as 100 viable juveniles per 100 cc of soil. Field experience shows that one year under a non-host crop (maize, sorghum or a marigold trap crop) reduces the viable population to about 40 per cent of the previous value, while one year under a susceptible host (tomato) multiplies it about three times.

1 **Plan A (continuous susceptible crop).** Year 1: 100 x 3 = 300. Year 2: 300 x 3 = 900. Year 3: 900 x 3 = 2,700. The population rises sharply - rotation fails.

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2 **Plan B (rotation).** Year 1 under maize (non-host): 100 x 0.40 = **40**. Year 2 under sorghum (non-host): 40 x 0.40 = **16**. Year 3 under a marigold trap crop plus a root-knot-resistant tomato: the population is kept low, say 16 x 0.40 = **6.4**.

3 Reduction over the three years in Plan B = (100 - 6.4) / 100 x 100 = **about 94 per cent**.

4 Conclusion: a three-year rotation of non-host crops (maize, sorghum, marigold) with a resistant variety in the host year brings the population down from 100 to about 6 per 100 cc - well below the damage threshold - whereas continuous tomato makes the problem three times worse every year.

The multiplication and reduction factors (3x for a host, 0.4 for a non-host) are illustrative. In practice they depend on soil, temperature, nematode species and crop; use locally determined values where available.

14.5 Management of Nematodes in These Crops

- Rotate with non-host cereals (maize, sorghum) and avoid susceptible crops such as castor in cotton rotations.
- Grow resistant or tolerant varieties of pulses, soybean and cotton where available.
- Treat seed with bioagents - *Pseudomonas fluorescens* and *Trichoderma viride* - before sowing.
- Apply organic amendments (neem cake, castor cake) and well-decomposed farmyard manure.
- Apply *Paecilomyces lilacinus* to the soil to parasitise root-knot eggs.
- Manage the wilt complex together - control both the nematode and the *Fusarium* or *Ralstonia* pathogen.
- Apply need-based nematicides (carbofuran 3G or phorate 10G at about 1-2 kg a.i./ha) where the population is above the threshold.

14.6 Managing the Wilt Complex in Pulses

In pulses the nematode is often only half of the problem; the other half is a wilt pathogen such as Fusarium or Ralstonia. Effective management must deal with both:

- Use varieties that resist both the nematode and the wilt pathogen where available.
- Treat seed with Pseudomonas fluorescens, which suppresses both the nematode and the wilt fungus.
- Rotate with non-host cereals to reduce the nematode, and choose a rotation that also reduces the wilt pathogen.
- Apply organic amendments and bioagents (Trichoderma, Paecilomyces lilacinus) to the soil.
- Remove and destroy wilted plants promptly, and avoid moving infested soil between fields.
- Use a need-based nematicide only where the population is high, and combine it with the other measures.

Exam focus - Lecture 14

Key points. (1) Pigeonpea is attacked by the cyst nematode Heterodera cajani; chickpea, blackgram and greengram mainly by Meloidogyne. (2) Soybean cyst nematode is Heterodera glycines. (3) Groundnut is attacked by Meloidogyne arenaria and Pratylenchus. (4) Cotton is attacked mainly by

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the reniform nematode Rotylenchulus reniformis, and Meloidogyne forms a wilt complex. (5) Jute is attacked by Meloidogyne and Hoplolaimus. (6) Crop rotation with non-host cereals reduces the population sharply, while continuous susceptible cropping multiplies it. (7) Manage the nematode together with the associated wilt pathogens.

Short-answer questions.

Q1. Name the cyst nematode of pigeonpea and the cyst nematode of soybean. *Model answer:* Pigeonpea - Heterodera cajani; soybean - Heterodera glycines.

Q2. Which nematode is most important on cotton in India and why is castor avoided in its rotation? *Model answer:* Rotylenchulus reniformis (reniform nematode); castor is a good host of the same nematode, so it would increase the population rather than reduce it.

Q3. Give two ways to reduce a root-knot population by crop rotation. *Model answer:* Grow non-host cereals such as maize or sorghum, and include a marigold trap crop; each non-host year reduces the viable population substantially.

Long-answer questions.

Q1. Describe the nematode problems of pulses and their management. *Outline:* pigeonpea (Heterodera cajani) - chickpea - blackgram and greengram - cowpea and soybean - symptoms and wilt complexes - management: rotation, resistant varieties, bioagents, organic amendments, need-based nematicides.

Q2. Discuss nematodes of oilseed and fibre crops and explain the use of crop rotation with a worked calculation. *Outline:* groundnut, sunflower, mustard, sesame/castor - cotton (Rotylenchulus) and jute - symptoms - rotation plan and population calculation - factors affecting the rate of decline - management package.

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Lecture 15: Nematode Parasites of Fruit and Plantation Crops

15.1 Introduction

Fruit and plantation crops are perennial, so once a nematode is introduced it multiplies season after season and is very hard to remove. Damage shows up as a slow decline in vigour and yield rather than as a sudden loss. In India the most important problems are the burrowing nematode of banana, the citrus nematode of citrus, and the virus-vectoring nematodes of grapevine.

15.2 Banana

- *Radopholus similis* - the burrowing nematode. It tunnels through the root cortex and the corm, leaving dark lesions and cavities, and destroys the root system. The classic result is "toppling disease" (blackhead toppling), in which the plant falls over because the roots can no longer hold it. It is the most important nematode of banana in India.
- *Meloidogyne* spp. - root galls on banana.
- *Helicotylenchus multicinctus* - the spiral nematode; causes root lesions and decline.
- *Pratylenchus* spp. - root lesions.

Management: use nematode-free suckers; pare the corms and give a hot-water treatment (about 50-55 degrees C for a short period) to kill nematodes in the planting material; apply bioagents and organic amendments; apply need-based nematicides.

15.3 Citrus

- *Tylenchulus semipenetrans* - the citrus nematode, causing "citrus slow decline". The tree slowly loses vigour over several years, with sparse yellow foliage, dieback and reduced fruit size and yield.
- *Radopholus citrophilus* (*R. similis*) - causes "spreading decline", a more rapid decline in localised patches, mainly reported from Florida; the burrowing nematode is also present on citrus in India.
- *Pratylenchus* spp. - root lesions.

Management: use resistant rootstocks and certified nursery stock; apply organic amendments and bioagents; manage the soil around the tree; use need-based nematicides in the basin.

15.4 Coconut

- In the Americas and the Caribbean, *Rhadinaphelenchus* (*Bursaphelenchus*) *cocophilus* causes red ring disease of coconut, in which the trunk shows a reddish ring and the palm dies.
- In India the important nematodes of coconut are *Meloidogyne*, *Helicotylenchus*, *Pratylenchus* and *Radopholus*; they weaken the root system and contribute to poor vigour and yield.

Red ring disease is not a major problem of coconut in India; it is largely a disease of the Neotropical region. Indian coconut nematode problems are mainly root-associated.

15.5 Tea, Coffee and Arecanut

- Tea: *Meloidogyne*, *Pratylenchus* and *Radopholus* attack the roots, causing decline and reduced yield.
- Coffee: *Meloidogyne* (root-knot) and *Pratylenchus* cause root damage and decline; *Radopholus* also occurs.
- Arecanut: *Meloidogyne*, *Radopholus* and *Helicotylenchus* cause root damage and decline in areca gardens.

15.6 Grapevine

Grapevine is attacked by the dagger nematode *Xiphinema index*, which is also the vector of grapevine fanleaf virus. The disease - fanleaf - causes malformed, fan-shaped leaves, yellow mottling and poor fruit set, and can destroy a vineyard. Because the virus is carried by the nematode, controlling the nematode is essential to control the disease.

15.7 Other Fruit Crops

- Mango: *Rotylenchulus reniformis*, *Hoplolaimus* and *Meloidogyne* are associated with decline and poor vigour.
- Papaya: *Meloidogyne* and *Rotylenchulus* cause stunting and poor fruit yield; papaya is very susceptible to root-knot.
- Pomegranate: *Meloidogyne* causes root galls and decline.
- Guava: *Meloidogyne* and *Helicotylenchus* are associated with decline.
- Apple and peach: *Pratylenchus* and *Meloidogyne* cause root damage; peach is a good host of root-knot.

15.8 Plantation and Other Field Crops

- Sugarcane: *Pratylenchus*, *Meloidogyne* and *Hoplolaimus* cause root damage and poor growth.
- Cotton (as a plantation-style crop in some regions): *Rotylenchulus reniformis* and *Meloidogyne*.
- Rubber: root-associated nematodes contribute to decline in some situations.

Crop	Main nematodes	Symptoms
Banana	Radopholus similis, Meloidogyne, Helicotylenchus	Root lesions, cavities, toppling disease
Citrus	Tylenchulus semipenetrans, Radopholus	Slow decline, sparse foliage, dieback
Coconut	Meloidogyne, Helicotylenchus, Pratylenchus	Poor root system, decline (red ring only in the Americas)
Tea, coffee	Meloidogyne, Pratylenchus, Radopholus	Root damage, decline, low yield
Grapevine	Xiphinema index	Fanleaf virus disease (mottling, malformed leaves)
Mango	Rotylenchulus, Hoplolaimus, Meloidogyne	Decline, poor vigour

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Crop	Main nematodes	Symptoms
Papaya, pomegranate	Meloidogyne, Rotylenchulus	Stunting, galls, poor yield
Sugarcane	Pratylenchus, Meloidogyne, Hoplolaimus	Root damage, poor growth

15.9 Management of Nematodes in Fruit and Plantation Crops

- Use nematode-free planting material - certified suckers, seedlings and rootstocks.
- Treat planting material: pare and hot-water-treat banana suckers; use resistant citrus rootstocks.
- Apply organic amendments (neem cake, farmyard manure) and bioagents (*Pseudomonas fluorescens*, *Trichoderma viride*, *Paecilomyces lilacinus*) to the soil around the plant.
- Maintain soil health with cover crops and mulching; avoid water stress, which worsens nematode damage.
- Apply need-based nematicides in the root zone (basin) of perennial crops; because the crop is perennial, chemical control is costly and must be justified by the population level.
- In grapevine, control *Xiphinema index* to prevent the spread of fanleaf virus.

15.10 Why Perennial Crops Are Special

Nematode management in fruit and plantation crops differs from that in annual crops in several ways:

- The crop stays in the field for years, so the nematode population builds up steadily and cannot be reset by ploughing or rotation.
- The whole field cannot be rotated; only the inter-space or cover crop can be changed.
- The planting material (suckers, seedlings, rootstocks) is the main way nematodes enter a new plantation, so clean planting material is critical.
- Chemical control is costly because it must cover the root zone of large plants, and it must be repeated.
- Prevention - clean material, resistant rootstocks, hot-water treatment and soil health - is therefore far more important than cure.

Exam focus - Lecture 15

Key points. (1) Radopholus similis causes toppling disease of banana by destroying the root system. (2) Tylenchulus semipenetrans causes citrus slow decline; Radopholus causes spreading decline. (3) In India, coconut nematodes are root-associated (Meloidogyne, Helicotylenchus, Pratylenchus); red ring disease is largely a Neotropical problem. (4) Xiphinema index transmits grapevine fanleaf virus. (5) Tea and coffee suffer from Meloidogyne, Pratylenchus and Radopholus. (6) Management relies on clean planting material, hot-water treatment of suckers, resistant rootstocks, organic amendments, bioagents and need-based nematicides.

Short-answer questions.

Q1. Which nematode causes toppling disease of banana and how? *Model answer:* Radopholus similis, the burrowing nematode; it tunnels through the root cortex and corm, destroys the root

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system, and the plant topples over.

Q2. What is citrus slow decline and which nematode causes it? *Model answer:* A gradual loss of vigour, sparse yellow foliage, dieback and reduced yield in citrus over several years, caused by Tylenchulus semipenetrans.

Q3. How is grapevine fanleaf virus spread? *Model answer:* By the dagger nematode Xiphinema index, which transmits the virus while feeding on the roots.

Long-answer questions.

Q1. Describe the nematode problems of banana and citrus with their management. *Outline:* banana - Radopholus similis and toppling - Meloidogyne and Helicotylenchus - management: clean suckers, paring and hot-water treatment, bioagents, nematicides. Citrus - Tylenchulus semipenetrans and slow decline - Radopholus and spreading decline - management: resistant rootstocks, certified stock, bioagents, basin treatment.

Q2. Discuss the nematodes of plantation crops and the special problems of perennial crops. *Outline:* banana, citrus, coconut, tea, coffee, grapevine, mango, papaya - symptoms - why perennials are harder to manage - clean planting material - hot-water treatment - resistant rootstocks - bioagents - need-based chemicals - virus vector control in grapevine.

Lecture 16: Principles of Nematode Management

16.1 The Basic Principle

Nematodes cannot usually be eradicated from a field once they are established, because they survive deep in the soil, inside roots and inside protective structures such as cysts and egg masses. The aim of nematode management is therefore not eradication but keeping the population below the damage threshold so that the crop does not suffer economic loss. This is why the word "management" is preferred to "control".

> *A key concept is the damage threshold (also called the economic threshold): the nematode population at which the value of the crop lost is greater than the cost of control. Management is justified only when the population is at or above this threshold.*

16.2 Objectives of Management

- Exclusion - keep the nematode out of clean fields (quarantine, clean seed and planting material).
- Eradication or reduction - reduce the population in the soil before planting.
- Protection - protect the plant during its vulnerable early stages.
- Resistance - use varieties that resist or tolerate the nematode.
- Avoidance - adjust the crop, season or field so that the nematode does not attack the crop.

16.3 Legislative (Regulatory) Methods

These methods use law and regulation to prevent the entry and spread of nematodes.

- Plant quarantine - restricts the movement of plants, seeds, soil and planting material across national borders and between states. In India this is governed by the Destructive Insects and Pests Act (1914), the Plant Quarantine Order (2003) and related rules.
- Certification of seed and planting material - only certified, nematode-free material is allowed to be sold or moved.
- Inspection and disinfestation of imported consignments.

Advantage: prevents the introduction of new, dangerous nematodes. Limitation: enforcement is difficult, and once a nematode enters a country, quarantine cannot remove it.

16.4 Physical Methods

- Soil solarisation - moist soil is covered with transparent polythene (about 25 micrometre thick) for 4-6 weeks in summer; the trapped heat (about 40-50 degrees C in the top soil) kills many nematodes.
- Heat treatment - hot-water treatment of bulbs, sets, suckers and other planting material (for example, banana suckers at about 50-55 degrees C for a short period).
- Steaming and dry heat - used in nursery and greenhouse soils.
- Flooding - prolonged flooding reduces many soil nematodes by creating low-oxygen conditions.
- Drying - drying of soil and planting material reduces nematode survival.
- Irradiation, microwaves and ultraviolet light - used in special situations.

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Advantage: no chemical residues; good for nurseries. Limitation: needs equipment, sun, water or time, and does not reach deep soil.

16.5 Cultural Methods

- Crop rotation with non-host crops - the single most useful cultural method; breaks the build-up of host-specific nematodes.
- Fallowing - keeping the field free of host plants for a period, so that the population declines.
- Organic amendments - neem cake, castor cake, mustard cake and farmyard manure release ammonia and other substances that reduce nematodes and also improve soil health.
- Trap crops - crops such as marigold (*Tagetes*) and *Crotalaria* that stimulate hatching but do not allow the nematode to complete its life cycle.
- Antagonistic crops - crops whose root exudates suppress nematodes.
- Crop sanitation - removing and destroying infected plants, crop residues and weeds.
- Time of planting - planting when soil temperature is unfavourable to the nematode.
- Mulching and soil tillage - help to disturb and expose nematodes to drying and heat.

Advantage: cheap, sustainable and compatible with other methods. Limitation: slow to act, and requires planning over seasons.

16.6 Biological Methods

Biological control uses living organisms that attack nematodes.

- *Paecilomyces lilacinus* (now *Purpureocillium lilacinum*) - a fungus that parasitises nematode eggs, especially those of root-knot nematodes.
- *Pochonia chlamydosporia* (*Verticillium chlamydosporium*) - another egg-parasitising fungus.
- *Trichoderma viride* and *T. harzianum* - fungi that compete with and antagonise nematodes and soil pathogens; widely recommended by TNAU.
- *Pseudomonas fluorescens* - a bacterium that reduces nematode infection and also controls many soil-borne diseases; a TNAU-recommended bioagent.
- *Pasteuria penetrans* - a bacterium that attaches to and kills the juveniles of root-knot nematodes.
- Nematode-trapping fungi such as *Arthrobotrys* - they capture nematodes with sticky or constricting traps.
- Predators - tardigrades, mites and predatory nematodes.

Advantage: safe, eco-friendly, and can be integrated with other methods. Limitation: slower than chemicals, sensitive to soil conditions, and must be applied before the population is high.

16.7 Chemical Methods (Nematicides)

Nematicides are chemicals that kill or immobilise nematodes. They are grouped as fumigants and non-fumigants.

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16.7.1 Fumigants

These are volatile liquids or gases that spread through the soil as a vapour and kill nematodes in the soil pores. Examples: methyl bromide, 1,3-dichloropropene (DD), chloropicrin, metam sodium and dazomet. Fumigation gives good control but is costly, needs careful handling, and methyl bromide has been largely phased out because it damages the ozone layer.

16.7.2 Non-fumigants

These are applied to the soil or as seed treatment and are taken up by the root or act on contact. Examples: carbamates (carbofuran, aldicarb, oxamyl, carbosulfan) and organophosphates (phorate, fenamiphos, ethoprophos).

16.7.3 Common nematicides and their doses

Nematicide	Group	Method of use	Standard dose range
Carbofuran 3G	Carbamate	Soil application	about 1-2 kg a.i./ha
Phorate 10G	Organophosphate	Soil application	about 1-2 kg a.i./ha
Aldicarb 10G	Carbamate	Soil application	about 1-2 kg a.i./ha (highly toxic)
Fenamiphos 10G	Organophosphate	Soil application	about 1-2 kg a.i./ha
Carbosulfan 25 EC	Carbamate	Seed treatment / soil	as per label recommendation
Methyl bromide	Fumigant	Soil fumigation	about 500-1,000 kg/ha (largely phased out)
1,3-dichloropropene (DD)	Fumigant	Soil fumigation	about 200-400 L/ha
Metam sodium	Fumigant	Soil drench / injection	about 500-1,000 L/ha
Dazomet	Fumigant (granular)	Soil incorporation	about 300-600 kg/ha

Many older nematicides (for example aldicarb and methyl bromide) are highly toxic or restricted, and some have been withdrawn. Safer options and bioagents are preferred in integrated management.

16.7.4 Drawbacks of nematicides

- Highly toxic to humans and animals; require protective equipment.
- Leave residues in soil, water and produce.
- Harmful to non-target and beneficial soil organisms.
- Costly, and repeated use is not sustainable.
- Continuous use may lead to resistance or to a rebound in the population.

16.8 Host-Plant Resistance

This is the cheapest and most farmer-friendly method. A resistant plant does not allow the nematode to feed or reproduce normally, while a tolerant plant is attacked but still yields well.

- Resistance is usually due to a pre-infection mechanism (the nematode cannot enter) or a post-infection mechanism (the nematode enters but cannot form feeding cells, so it dies or fails to reproduce).
- Examples: tomato varieties carrying the *Mi* gene resist root-knot nematodes; resistant varieties of wheat and barley resist the cereal cyst nematode; resistant soybean varieties resist the soybean cyst nematode.

- Resistance can be broken by high soil temperature or by a mixed population of nematode species, so it should be used with other methods.

Advantage: low cost, no residues, easy for farmers. Limitation: not available for every crop and nematode, and resistance may be overcome.

16.9 Comparison of Management Methods

Method	Advantages	Limitations
Legislative (quarantine)	Prevents entry of new nematodes; protects clean areas	Hard to enforce; cannot remove a nematode once it enters
Physical (solarisation, heat, flooding)	No residues; good for nurseries; effective in the top soil	Needs equipment, sun, water or time; does not reach deep soil
Cultural (rotation, organic matter, trap crops)	Cheap; sustainable; improves soil health	Slow; needs advance planning over seasons
Biological (P. lilacinus, Trichoderma, P. fluorescens, Pasteuria)	Eco-friendly; integrates with other methods	Slower than chemicals; sensitive to soil conditions
Chemical (nematicides)	Fast, reliable knockdown of high populations	Toxic, costly, residues, harmful to non-targets, some withdrawn
Host-plant resistance	Cheapest; no residues; easy to adopt	Not available for all crops; may break down

Worked example: estimating a nematode population from a soil sample

A composite soil sample of 100 cc is processed by sieving and decanting followed by a Baermann funnel, and the nematodes are collected in a final suspension of 100 mL. Counting under the microscope shows an average of 3 second-stage juveniles (J2) of *Meloidogyne* per 1 mL aliquot.

1. Total nematodes in the suspension = 3 J2/mL x 100 mL = **300 J2**.
2. These 300 J2 came from 100 cc of soil, so the population is **300 J2 per 100 cc of soil**.
3. If 100 cc of this soil weighs about 130 g, the population per gram = 300 / 130 = **about 2.3 J2 per gram of soil**.
4. The action threshold for root-knot on tomato is about 2 J2 per gram of soil. Since 2.3 is above 2.0, **management is justified** - for example, a bioagent such as Paecilomyces lilacinus, or a need-based nematicide such as carbofuran 3G at about 1-2 kg a.i./ha.
5. If the initial population were Pi = 2.3 J2/g and, after the crop, the final population were Pf = 9.2 J2/g, the reproduction factor Rf = Pf / Pi = 9.2 / 2.3 = **4.0**, showing that the crop was a good host (Rf greater than 1 means the population increased).

Always state the volume of soil processed, the final suspension volume and the aliquot counted, because these decide the multiplication factor.

16.10 Safe Use of Nematicides

Nematicides are among the most toxic of all pesticides, so safety is essential:

- Read and follow the label and the state recommendation; never exceed the dose.
- Wear protective clothing, gloves, a mask and eye protection during handling and application.
- Do not eat, drink or smoke while handling nematicides; wash thoroughly afterwards.
- Store chemicals away from food, feed and children; dispose of empty containers safely.
- Observe the waiting period between application and harvest.
- Prefer bioagents and cultural methods where they can do the job, to reduce the chemical load.

16.11 Cost-Benefit of Management

Management is justified only when the value of the crop saved is more than the cost of the treatment. The decision follows from the damage threshold: estimate the likely loss from the current population, estimate the cost of the chosen method, and act only if the expected saving is greater. A cheap method such as crop rotation or a bioagent often gives a better return than an expensive fumigant, which is why integrated management is preferred.

Exam focus - Lecture 16

Key points. (1) Nematodes are managed, not eradicated; the aim is to keep the population below the damage threshold. (2) The main approaches are exclusion, eradication/reduction, protection, resistance and avoidance. (3) Legislative methods use quarantine and certification to keep nematodes out. (4) Physical methods include solarisation, heat treatment, flooding and drying. (5) Cultural methods include crop rotation, organic amendments, trap crops and sanitation. (6) Biological control uses P. lilacinus, Pochonia, Trichoderma, P. fluorescens and Pasteuria penetrans. (7) Nematicides are fumigants (methyl bromide, DD, metam sodium, dazomet) and non-fumigants (carbofuran, phorate, aldicarb, fenamiphos), usually at about 1-2 kg a.i./ha. (8) Host-plant resistance is the cheapest and safest method.

Short-answer questions.

Q1. Why is the term "management" preferred to "control" for nematodes? *Model answer:* Because nematodes cannot usually be eradicated from a field; the practical aim is to keep the population below the damage threshold, which is management rather than complete control.

Q2. Name any four bioagents used against nematodes. *Model answer:* Paecilomyces lilacinus, Pochonia chlamydosporia, Trichoderma viride and Pseudomonas fluorescens (also Pasteuria penetrans).

Q3. Distinguish a resistant plant from a tolerant plant. *Model answer:* A resistant plant does not allow the nematode to feed or reproduce normally; a tolerant plant is attacked but still produces a good yield.

Long-answer questions.

Q1. Describe the different methods of nematode management with their advantages and limitations. *Outline:* principle of management and the damage threshold - legislative - physical - cultural - biological - chemical - host-plant resistance - comparison table.

Q2. Explain chemical control of nematodes and its drawbacks. *Outline:* fumigants and non-fumigants

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- examples and dose ranges - methods of application (soil, seed, nursery) - advantages - drawbacks (toxicity, residues, non-target effects, cost, resistance) - place of chemicals in integrated management - worked population example.

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Lecture 17: Integrated Nematode Management

17.1 The Concept of Integrated Nematode Management

Integrated nematode management (INM), also called integrated disease management (IDM) for nematodes, is the combined use of all suitable methods - legislative, physical, cultural, biological, chemical and host-plant resistance - in a planned way, so that the nematode population is kept below the damage threshold with the least possible use of chemicals. No single method is enough on its own; INM brings them together so that each method supports the others.

The guiding ideas are:

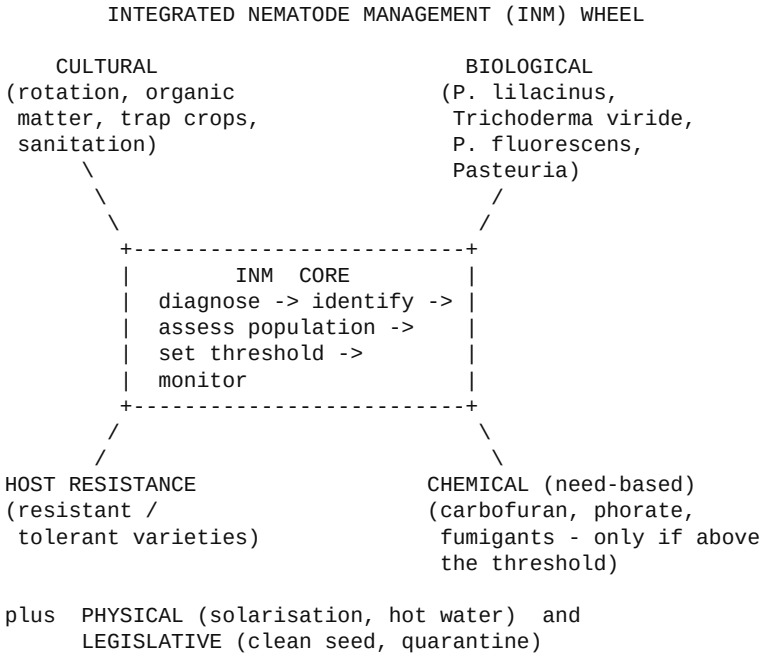
- Prevention first - stop the nematode from entering the field or from building up.
- Combine methods - rotation, resistance and bioagents do most of the work; chemicals are used only when needed.
- Keep it economical - the cost of management must be less than the value of the crop saved.
- Keep it safe - protect human health, beneficial organisms and the environment.

17.2 Steps in Developing an INM Programme

1. Diagnose the problem - confirm that nematodes, not something else, are causing the damage (patchy growth, galls, cysts, poor roots).
2. Identify the nematode - find out the species or genus, because management depends on it.
3. Assess the population - estimate the nematode population from a soil and root sample (see the worked example in Lecture 16).
4. Set the threshold - compare the population with the damage threshold for the crop.
5. Choose and combine methods - select the cheapest effective combination of cultural, biological, resistant and (if needed) chemical methods.
6. Monitor and review - sample again after the crop to see whether the population has fallen; adjust the plan for the next season.

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17.3 The Integrated Management Wheel



17.4 Crop-wise Integrated Management Packages

17.4.1 Rice (upland root-knot and other rice nematodes)

- Use clean, nematode-free seed (important for white tip).
- Grow tolerant or resistant varieties where available.
- Rotate or follow with a non-host crop; where possible, flood the field, which reduces many soil nematodes.
- Apply neem cake and well-decomposed organic matter.
- Treat seed or nursery with *Pseudomonas fluorescens* and *Trichoderma viride*.
- Apply a need-based nematicide (carbofuran 3G at about 1-2 kg a.i./ha) only if the population is above the threshold.

17.4.2 Tomato (root-knot nematode)

- Grow a root-knot-resistant variety (for example, one carrying the *Mi* gene).
- Solarise the nursery bed and use treated seedlings.
- Include marigold (*Tagetes*) as a trap crop or intercrop.
- Apply neem cake and farmyard manure.
- Apply *Paecilomyces lilacinus*, *Trichoderma viride* and *Pseudomonas fluorescens* to the soil or as seedling treatment.
- Use a need-based nematicide only when the gall index or soil population is above the threshold.

17.4.3 Banana (burrowing nematode)

- Use nematode-free suckers; pare the corms and give a hot-water treatment.
- Apply organic amendments and bioagents in the planting pit.
- Maintain good soil moisture and drainage; avoid stress.
- Apply a need-based nematicide in the root zone if the population is high.

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17.4.4 Pulses (cyst and root-knot nematodes)

- Rotate with non-host cereals such as maize and sorghum.
- Grow resistant or tolerant varieties where available.
- Treat seed with *Pseudomonas fluorescens* and *Trichoderma viride*.
- Manage the wilt complex together with the *Fusarium* or *Ralstonia* pathogen.
- Use a need-based nematicide only where justified.

17.5 Typical TNAU-Style Recommendations

Practice	Typical recommendation
Pseudomonas fluorescens	Seed treatment (about 10 g/kg seed) or soil application (about 2.5 kg/ha with organic matter)
Trichoderma viride	About 2 kg/ha, applied with farmyard manure
Paecilomyces lilacinus	About 2.5 kg/ha, applied to the soil to parasitise root-knot eggs
Neem cake	About 250-500 kg/ha (up to about 1 t/ha), incorporated before planting
Soil solarisation	Transparent polythene (about 25 micrometre) for 4-6 weeks in summer
Carbofuran 3G	About 1-2 kg a.i./ha in the soil, only when the population is above the threshold
Marigold (Tagetes)	Grown as a trap or intercrop to reduce root-knot populations

17.6 Benefits of Integrated Nematode Management

- Reduces the nematode population steadily over seasons.
- Cuts the cost of control by reducing dependence on nematicides.
- Protects human health and the environment by using fewer chemicals.
- Preserves beneficial soil organisms and improves soil health.
- Gives more stable yields, because several methods support one another.
- Slows the development of resistance to chemicals and of breakdown of host resistance.

17.7 An Example of an INM Plan for Tomato

1. Sample the field and find the root-knot population and gall index. Suppose the population is 2.3 J2 per gram of soil (above the threshold of 2).
2. Choose a root-knot-resistant tomato variety.
3. Solarise the nursery bed and treat seedlings with *Pseudomonas fluorescens*.
4. Apply neem cake (about 250-500 kg/ha) and *Paecilomyces lilacinus* (about 2.5 kg/ha) to the main field before planting.
5. Plant marigold as an intercrop to trap the nematode.
6. Because the population is only slightly above the threshold, avoid a nematicide in the first season; instead monitor and sample again. If the population is much higher, apply carbofuran 3G at about 1-2 kg a.i./ha.
7. After harvest, sample again and record the fall in population; adjust the plan for the next crop.

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17.8 Monitoring and Record Keeping

A good INM programme is a cycle, not a one-time action. Keep simple records for each field:

- The nematode species and population before planting (per 100 cc or per gram of soil).
- The methods used (variety, rotation, bioagents, chemicals) and their dates.
- The root-gall index or cyst count at the end of the crop.
- The final population and the reproduction factor (Rf = Pf / Pi).
- The yield and the approximate loss avoided.

These records show whether the population is falling or rising, and they guide the next season's plan.

17.9 Common Mistakes in Integrated Management

- Using only one method (usually a chemical) and ignoring rotation and resistance.
- Applying a nematicide without checking the population - wasting money when the population is low.
- Growing the same susceptible crop year after year on the same land.
- Planting nematode-infested nursery stock or suckers into clean soil.
- Forgetting the wilt complex - controlling the nematode but not the fungus or bacterium.
- Applying bioagents after the population is already very high, when they act too slowly.
- Ignoring sanitation - leaving infected crop debris and weeds in the field.

Exam focus - Lecture 17

Key points. (1) Integrated nematode management combines legislative, physical, cultural, biological, chemical and host-resistance methods to keep the population below the threshold. (2) The steps are diagnose, identify, assess population, set threshold, combine methods, and monitor. (3) Prevention and combination are the guiding ideas; chemicals are used only when needed. (4) A typical package uses resistant varieties, rotation, organic amendments, bioagents (P. fluorescens, T. viride, P. lilacinus) and need-based nematicides. (5) Crop-wise packages are available for rice, tomato, banana and pulses. (6) INM reduces cost, protects the environment and gives stable yields.

Short-answer questions.

Q1. Define integrated nematode management. *Model answer:* The combined and planned use of all suitable methods - legislative, physical, cultural, biological, chemical and host resistance - to keep the nematode population below the damage threshold with the least use of chemicals.

Q2. List the steps in developing an INM programme. *Model answer:* Diagnose the problem, identify the nematode, assess the population, set the threshold, combine suitable methods, and monitor and review after the crop.

Q3. Give any three benefits of INM. *Model answer:* Lower control cost, less chemical pollution and harm to beneficial organisms, and more stable yields over seasons (also slows resistance development).

Long-answer questions.

Q1. Explain the concept and components of integrated nematode management with a suitable diagram. *Outline:* definition and guiding ideas - steps (diagnose, identify, assess, threshold, combine, monitor) - components (legislative, physical, cultural, biological, chemical, resistance) - the INM wheel diagram - benefits.

Q2. Describe an integrated management package for tomato or rice. *Outline:* identify the main nematode - resistant variety - nursery and seed treatment - solarisation - organic amendments - bioagents (P. fluorescens, T. viride, P. lilacinus) - trap crop - need-based nematicide - monitoring - expected outcome.

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Appendix A: Common Names of Important Plant-Parasitic Nematodes

A.1 Why Learn Common Names

Common names are widely used in field advisory work and in examinations. Each common name usually describes the feeding organ or the symptom, so learning the name helps to remember the biology. The table below lists the common names of the important plant-parasitic nematode genera and species.

Common name	Scientific name	Note
Root-knot nematode	Meloidogyne spp.	Galls on roots; very wide host range
Cyst nematode	Heterodera spp.	Lemon-shaped cysts; syncytia
Cereal cyst nematode	Heterodera avenae	"Molya" of wheat
Pigeonpea cyst nematode	Heterodera cajani	Cyst nematode of pigeonpea
Soybean cyst nematode	Heterodera glycines	Major soybean problem
Potato cyst nematode	Globodera rostochiensis, G. pallida	Golden and pale cyst nematodes
Root-lesion nematode	Pratylenchus spp.	Migratory endoparasite; root lesions
Burrowing nematode	Radopholus similis	Banana toppling; citrus spreading decline
Citrus nematode	Tylenchulus semipenetrans	Citrus slow decline; semi-endoparasite
Reniform nematode	Rotylenchulus reniformis	Kidney-shaped female; wide host range
Dagger nematode	Xiphinema spp.	Ectoparasite; vector of Nepoviruses
Needle nematode	Longidorus spp.	Ectoparasite; vector of Nepoviruses
Stubby-root nematode	Trichodorus, Paratrichodorus	Ectoparasite; vector of Tobraviruses
Sting nematode	Belonolaimus spp.	Ectoparasite; severe on sandy soils
Lance nematode	Hoplolaimus spp.	Semi-endoparasite; root damage
Spiral nematode	Helicotylenchus spp.	Very common in soil; root damage
Ring nematode	Criconemoides (Mesocriconema) spp.	Ectoparasite; ringed cuticle
Sheath nematode	Hemicycliophora spp.	Ectoparasite; retains old cuticle
Pin nematode	Paratylenchus spp.	Small ectoparasite
Stem and bulb nematode	Ditylenchus dipsaci	Onion bloat; "eelworm wool"
Seed-gall / ear-cockle nematode	Anguina tritici	Historic wheat nematode
Foliar nematode	Aphelenchoides spp.	Leaves and buds
White-tip nematode	Aphelenchoides besseyi	White tip of rice; seed-borne
Rice root nematode	Hirschmanniella oryzae, H. mucronata	Wetland rice
Ufra nematode	Ditylenchus angustus	Ufra disease of rice
Root-knot nematode of rice	Meloidogyne graminicola	Upland rice and nurseries

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Common name	Scientific name	Note
Red ring nematode	Bursaphelenchus (Rhadinaphelenchus) cocophilus	Coconut, mainly Neotropics
Pine wood nematode	Bursaphelenchus xylophilus	Wilt of pine; forestry pest
False root-knot nematode	Nacobbus spp.	Galls like root-knot

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Appendix B: Symptom-Based Diagnostic Key

B.1 Using Symptoms to Reach a Nematode Group

A diagnostic key is a step-by-step guide that helps to narrow down the possible cause from the symptom. The key below uses only the symptoms seen in the field, so it can be used before any laboratory test. Always confirm with a soil and root sample, because several disorders look alike.

B.1.1 Below-ground symptoms

1 Swellings (galls or knots) on the roots: suspect *Meloidogyne* (root-knot nematode). Confirm by cutting a gall and looking for the pear-shaped female or egg mass.

2 Small lemon-shaped or round bodies on the roots, later free in the soil: suspect *Heterodera* or *Globodera* (cyst nematodes).

3 Dark, sunken lesions on the roots, with brown discolouration: suspect *Pratylenchus* (root-lesion nematode).

4 Cavities or tunnels inside the root cortex, roots breaking easily: suspect *Radopholus* (burrowing nematode).

5 Stubby, shortened roots with necrotic tips, no galls: suspect an ectoparasite such as *Trichodorus*, *Belonolaimus* or *Criconemoides*.

6 The posterior end of the female visible on the root surface with an egg mass, anterior end buried: suspect *Tylenchulus* or *Rotylenchulus* (semi-endoparasites).

B.1.2 Above-ground symptoms

1 Stunted, pale plants in patches, worse on sandy soil, with galls on roots: root-knot nematode.

2 Stunted, pale plants in patches in a cereal field, with cysts on roots: cereal cyst nematode ("molya").

3 Tip of the leaf white and twisted, small panicle, in rice: white tip (*Aphelenchoides besseyi*).

4 Swollen, distorted stem or bulb with rot, in onion or garlic: *Ditylenchus dipsaci*.

5 Galls in place of grain in wheat: *Anguina tritici* (ear cockle).

6 Slow, progressive loss of vigour over years in citrus: *Tylenchulus semipenetrans* (slow decline).

7 Malformed, fan-shaped leaves and mottling in grapevine: fanleaf virus, spread by *Xiphinema*.

8 Plants that wilt even when the soil is moist, with root galls: root-knot nematode, often with a wilt pathogen.

B.2 Caution

Nematode symptoms closely resemble those of nutrient deficiency, drought, waterlogging, root-rot fungi and soil compaction. The two most useful field clues are (1) patchiness of the damage and (2) a poor, galled or lesioned root system under a plant that looks starved. Always confirm by extracting and counting nematodes from a soil and root sample.

If you see	Most likely nematode
Galls (knots) on roots	Meloidogyne
Cysts on roots	Heterodera, Globodera
Dark lesions on roots	Pratylenchus
Cavities inside roots	Radopholus
Stubby roots, no galls	Trichodorus, Belonolaimus, Criconemoides
White leaf tip in rice	Aphelenchoides besseyi
Swollen onion bulb	Ditylenchus dipsaci
Galls in wheat grain	Anguina tritici
Slow decline of citrus	Tylenchulus semipenetrans
Fanleaf of grapevine	Xiphinema (virus vector)

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Appendix C: Laboratory Methods in Nematology

C.1 Introduction

Practical work is an essential part of nematology. Because nematodes are small and live in soil and roots, special methods are needed to sample, extract, count and identify them. This appendix summarises the standard methods that support the theory chapters.

C.2 Soil and Root Sampling

- Take a composite sample from many spots (10-20) in a zig-zag pattern, from the root zone at about 15-20 cm depth.
- Mix the soil thoroughly and take a sub-sample of about 100-250 cc for extraction.
- For root samples, lift plants with the root system intact, wash gently, and examine fresh; keep a part for staining.
- Sample when the soil is moist but not waterlogged, and keep samples cool and shaded until processing.

Sampling is the weakest link in nematode diagnosis. A poor sample gives a wrong answer even if the laboratory work is perfect, so collect many small sub-samples rather than one large one.

C.3 Extraction of Nematodes from Soil

Method	Principle	Best for
Cobb's sieving and decanting	Soil is washed through a series of sieves; nematodes are separated by size and settling	General soil nematodes, including large ones
Baermann funnel	Active nematodes migrate out of soil or tissue into water and settle at the bottom	Active nematodes from soil and plant tissue
Centrifugal flotation	Nematodes float in a dense sugar or salt solution and are separated by centrifugation	Cysts and soil nematodes; rapid
Seinhorst elutriator	A stream of water separates nematodes from soil by differential settling	Large soil samples
Mistifier / spray chamber	Nematodes migrate out of leaves and buds under a fine water spray	Foliar nematodes (Aphelenchoides)

C.4 Counting

- Collect the extracted nematodes in a known volume (for example, 100 mL) of water.
- Mix well, take a known aliquot (for example, 1 mL) with a counting dish or a Hawksley slide, and count under a stereomicroscope.
- Multiply by the dilution factor to get the population in the original soil volume (worked example in Lecture 16).
- Express the population as numbers per 100 cc of soil, or per gram of soil, and record the genus or species.

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C.5 Staining and Examining Roots

- Acid fuchsin staining - roots are boiled in acid fuchsin-lactophenol (or cleared in sodium hypochlorite and stained) so that nematodes inside the root appear red against the cleared tissue.
- Root-gall index - roots are graded 0-5 by the number of galls (see Lecture 13).
- Sectioning and squash mounts - to see giant cells, syncytia and the position of the female.

C.6 Preservation and Slide Making

- Nematodes are killed by gentle heat and fixed in TAF (triethanolamine-formalin) or in about 4 per cent formalin.
- For permanent slides they are processed through glycerine (Seinhorst or Cobb methods) and mounted.

C.7 Identification

- Morphological - study the cuticle, stylet, oesophagus, amphids, phasmids, tail and reproductive organs.
- Morphometric - record de Man's indices (L, W, a, b, c, c', V, T) and compare with published values.
- Molecular - extract DNA and use PCR and sequencing of rRNA genes to confirm the species.

C.8 Field and Pot Experiments

- Bioassay - grow a susceptible indicator plant in the test soil and examine its roots for galls or cysts.
- Pot experiments - inoculate known numbers of nematodes onto test plants to study multiplication (reproduction factor, Rf) and to screen varieties for resistance.

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Appendix D: Abbreviations and Symbols

D.1 Commonly Used Abbreviations

Abbreviation	Full form
a.i.	Active ingredient
a, b, c, c', V, T	de Man's indices
cc	Cubic centimetre (millilitre)
cf.	Compare with
DNA	Deoxyribonucleic acid
EDB	Ethylene dibromide
ETL	Economic threshold level
FYM	Farmyard manure
ha	Hectare
ICAR	Indian Council of Agricultural Research
IDM / INM	Integrated disease management / integrated nematode management
IARI	Indian Agricultural Research Institute
IMP	International Meloidogyne Project
J1-J4	First to fourth juvenile stages
kg	Kilogram
q	Quintal
L	Body length (also litre in other contexts)
Nepovirus	Nematode-transmitted polyhedral virus
NSI	Nematological Society of India
ONTA	Organization of Nematologists of Tropical America
PCR	Polymerase chain reaction
Pf / Pi	Final population / initial population
Rf	Reproduction factor (Pf / Pi)
rRNA	Ribosomal ribonucleic acid
SON	Society of Nematologists
sp. / spp.	Species (singular / plural)
TAF	Triethanolamine-formalin (fixative)
TNAU	Tamil Nadu Agricultural University
Tobravirus	Tobacco rattle virus group

D.2 Symbols and Conventions

- Scientific names of genus and species are written in italics, the genus with a capital letter and the species with a small letter (for example, *Meloidogyne incognita*).

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- A superscript after a name (for example, *Heterodera glycines* Ichinohe) is the name of the author who described the species.
- "sp." is used when the species is not known, and "spp." when more than one species is meant.
- Population counts are given per 100 cc of soil or per gram of soil; the basis is always stated.

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Appendix E: Model Question Papers

E.1 Model Theory Paper - Mid Semester (Lectures 1-8)

Section A - Answer all questions (2 marks each)

1. Define nematology. Who is called the father of nematology?
2. Give the systematic position of plant-parasitic nematodes.
3. What is a pseudocoelom and what is its function?
4. Name the parts of the oesophageal region in the Tylenchida.
5. Distinguish between an amphid and a phasmid.
6. What are spicules and the gubernaculum?
7. Define a sedentary endoparasite with one example.
8. What are entomopathogenic nematodes? Give one example.

Section B - Answer any five (5 marks each)

1. Write a note on the historical development of nematology in India.
2. Describe the body wall of a nematode.
3. Explain the excretory system of Secernentea and Adenophorea.
4. Describe the life cycle of a nematode.
5. Compare Meloidogyne and Heterodera.
6. Classify plant-parasitic nematodes by feeding habit.

Section C - Answer any two (10 marks each)

1. Describe the general morphology of a nematode with a labelled diagram.
2. Discuss the economic importance of nematodes.
3. Give an account of the classification of plant-parasitic nematodes into Secernentea and Adenophorea.

E.2 Model Theory Paper - Final / Semester (Lectures 10-17)

Section A - Answer all questions (2 marks each)

1. Why is nematode damage patchy in a field?
2. What is a disease complex? Give one example.
3. Name two nematode vectors of plant viruses.
4. Which nematode causes molya of wheat?
5. Which nematode causes toppling disease of banana?
6. What is the root-gall index?
7. Distinguish a resistant plant from a tolerant plant.
8. Define integrated nematode management.

Section B - Answer any five (5 marks each)

1. Describe the symptoms of nematode damage in crop plants.
2. Explain the interaction between a nematode and a wilt fungus.

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3 Write a note on the nematodes of rice.

4 Describe the nematode problems of vegetable crops.

5 Explain the principles of biological control of nematodes.

6 Describe soil solarisation and its use.

Section C - Answer any two (10 marks each)

1 Discuss the role of nematodes in disease development with the disease triangle.

2 Describe the methods of nematode management with their advantages and limitations.

3 Explain integrated nematode management with a crop-wise example.

E.3 Answer-Writing Hints

- Read the question carefully and answer only what is asked.
- Start with a definition, then give points, then a diagram or table where asked.
- In numerical questions, write the formula, substitute the values, and state the answer with its unit.
- Use scientific names in italics, and give one example with every statement wherever possible.
- Draw neat labelled diagrams; a good diagram often earns more marks than extra sentences.
- Manage time - do not spend too long on a 2-mark question.

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Appendix F: Crop-wise Quick Reference

F.1 How to Use This Table

This table brings together, in one place, the most important nematode of each crop group, its main symptom, and the first management steps. It is meant for quick revision before an examination and as a field reminder. The management column gives the general approach; always follow the current state or TNAU recommendation for exact doses and varieties.

Crop	Key nematode	Main symptom	First management steps
Rice	Meloidogyne graminicola	Root galls, nursery failure	Clean seed, solarised nursery, rotation, flooding, neem cake, bioagents
Rice	Aphelenchoides besseyi	White tip, poor grain filling	Clean seed, hot-water seed treatment
Wheat	Heterodera avenae	Molya, patchy stunting, cysts	Resistant varieties, rotation with non-cereals, summer ploughing
Wheat	Anguina tritici	Ear cockle, galls in grain	Clean seed, seed cleaning, crop rotation
Maize, sorghum	Pratylenchus, Meloidogyne	Root damage, stunting	Rotation, organic matter, bioagents
Tomato	Meloidogyne, Rotylenchulus	Root galls, wilt complex	Resistant variety (Mi gene), solarised nursery, marigold, bioagents
Brinjal	Meloidogyne, Rotylenchulus	Galls, bacterial wilt complex	Resistant variety, bioagents, manage wilt pathogen together
Okra	Meloidogyne, Rotylenchulus	Galls, stunting	Resistant variety, rotation, bioagents
Chilli	Meloidogyne, Rotylenchulus	Stunting, low fruit yield	Nursery treatment, rotation, bioagents
Potato	Globodera, Meloidogyne	Cysts, tuber damage	Resistant varieties, rotation, clean seed tubers
Onion, garlic	Ditylenchus dipsaci	Onion bloat	Clean sets, hot-water treatment, rotation
Cucurbits	Meloidogyne	Root galls	Resistant rootstocks, rotation, bioagents
Pigeonpea	Heterodera cajani, Meloidogyne	Cysts, galls, wilt complex	Resistant varieties, rotation, seed treatment
Chickpea	Meloidogyne, Pratylenchus	Galls, wilt complex	Resistant varieties, bioagents, wilt management
Blackgram, greengram	Meloidogyne, Rotylenchulus	Galls, poor podding	Resistant varieties, rotation, seed treatment
Soybean	Heterodera glycines	Cysts, stunting	Resistant varieties, rotation
Groundnut	Meloidogyne arenaria	Galls, pod damage	Resistant varieties, rotation, organic matter
Sunflower	Meloidogyne, Rotylenchulus	Stunting, poor head	Rotation, bioagents
Mustard, rapeseed	Meloidogyne, Heterodera	Galls, cysts	Rotation with cereals, resistant varieties

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Crop	Key nematode	Main symptom	First management steps
Cotton	Rotylenchulus reniformis	Stunting, low boll yield	Rotation (avoid castor), resistant varieties
Jute	Meloidogyne, Hoplolaimus	Stunting, low fibre	Rotation, organic matter, bioagents
Banana	Radopholus similis	Toppling disease	Clean suckers, paring + hot water, bioagents, organic matter
Citrus	Tylenchulus semipenetrans	Slow decline	Resistant rootstocks, certified stock, bioagents
Coconut	Meloidogyne, Helicotylenchus	Poor roots, decline	Organic matter, bioagents, soil health
Tea, coffee	Meloidogyne, Pratylenchus	Decline, low yield	Organic matter, bioagents, shade and soil management
Grapevine	Xiphinema index	Fanleaf virus	Control the vector; certified planting material
Mango, papaya	Rotylenchulus, Meloidogyne	Decline, stunting	Organic matter, bioagents, resistant rootstocks
Sugarcane	Pratylenchus, Meloidogyne	Poor growth	Rotation, organic matter, bioagents

F.2 A General Rule for Every Crop

Whatever the crop, the order of thinking is the same: sample the soil, identify and count the nematode, compare with the threshold, then combine a resistant variety, a rotation, organic matter and a bioagent, and use a nematicide only if the population is still above the threshold. This is the practical meaning of integrated nematode management.

Glossary

Glossary of Terms

Term	Meaning
Adenophorea (Aphasmidia)	Class of nematodes without phasmids; have large amphids, a glandular renette excretory system and caudal glands
Amphid	Paired chemosensory organ on the head (lips) of a nematode; its shape is important in taxonomy
Anus	Opening at the posterior end through which undigested waste leaves the gut
Bacterivore	A nematode that feeds on bacteria
Caudal glands	Glands in the tail region of Adenophorea; absent in Secernentea
Cloaca	Common chamber in the male through which both faeces and sperm pass
Crop rotation	Growing a sequence of different crops in a field to break the build-up of a host-specific nematode
Cyst	Hardened, tanned body wall of a dead Heterodera or Globodera female that holds 200-600 eggs
Damage threshold	Nematode population at which the value of crop lost exceeds the cost of control
Deirid	Small sense organ on the lateral field of a nematode
de Man's indices	Standard ratios (L, W, a, b, c, c', V, T) used to describe nematode size and proportions
Dioecious	Having separate male and female individuals
Ecdysis	Moulting of the cuticle; occurs four times during development
Ectoparasite	Nematode that remains outside the root and feeds on its surface
Egg mass	Gelatinous matrix in which Meloidogyne lays its eggs
Endoparasite	Nematode that enters the root tissue to feed
Entomopathogenic nematode	Nematode that kills insects (Steinernema, Heterorhabditis); used as a biopesticide
Excretory pore	Opening through which the excretory system empties to the outside
Giant cell (nurse cell)	Large, multinucleate feeding cell induced by Meloidogyne in the root vascular tissue
Gubernaculum	Cuticular guide that steers the spicules during copulation
Hermaphrodite	An individual that produces both eggs and sperm (e.g., Caenorhabditis elegans)
Hypodermis	Living layer beneath the cuticle that secretes the cuticle and forms the longitudinal cords
Incisure	A longitudinal line marking the lateral field of a nematode
J1, J2, J3, J4	The four juvenile stages; the J2 is the infective stage in root-knot and cyst nematodes
Migratory endoparasite	Nematode that enters the root and keeps moving through the tissue (e.g., Pratylenchus, Radopholus)
Monodelphic	Female with a single ovary and uterus
Monorchic	Male with a single testis

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Term	Meaning
Nematicide	Chemical used to kill or immobilise nematodes
Nematode	Unsegmented, bilaterally symmetrical, round-bodied worm of the phylum Nematoda
Odontostyle	Hollow feeding spear of the Dorylaimida (e.g., Xiphinema), formed from a modified tooth
Oesophagus (pharynx)	Muscular tube that pumps food from the mouth to the intestine
Parthenogenesis	Development of an egg without fertilisation (common in Meloidogyne)
Phasmid	Paired glandular sense organ near the tail; present only in Secernentea
Pseudocoelom	Fluid-filled body cavity that acts as a hydrostatic skeleton
Renette	Glandular excretory organ of Adenophorea
Reproduction factor (Rf)	Final population divided by initial population (Pf/Pi); Rf greater than 1 means the population increased
Resistant plant	A plant that does not allow the nematode to feed or reproduce normally
Secernentea (Phasmidia)	Class of nematodes with phasmids; small amphids, tubular H-shaped excretory system; includes the Tylenchida
Sedentary endoparasite	Nematode that enters the root, becomes fixed and induces feeding cells (Meloidogyne, Heterodera)
Semi-endoparasite	Nematode with the anterior end inside the root and the posterior outside (Tylenchulus, Rotylenchulus)
Soil solarisation	Heating moist soil under transparent polythene to kill nematodes
Spicule	Paired cuticular, needle-like male organ used to transfer sperm
Stylet (spear)	Hollow feeding spear of the Tylenchida
Syncytium	Multinucleate feeding cell induced by cyst nematodes
Taxonomy	Science of identifying, naming and classifying organisms
Tolerant plant	A plant that is attacked by the nematode but still yields well
Trap crop	A crop (e.g., marigold) that stimulates nematode hatching but does not allow it to complete its life cycle
Vector	An organism that carries a pathogen from plant to plant; some nematodes transmit viruses
Vulva	Female reproductive opening on the ventral side

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Suggested Reading

The following standard nematology texts and course materials are recommended for further study. Students should consult the latest editions available in their library.

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- *Indian Journal of Nematology*, Nematological Society of India (back volumes).

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Note on Sources and Figures

Where a field figure such as a percentage loss, a population threshold or a nematicide dose is given in this book, it is a standard published range rather than a single exact value, because the true value depends on the crop, cultivar, soil type, temperature and nematode population. Students should treat these figures as representative and follow the current TNAU or state recommendation for any actual field decision. Dates of a few early historical events and of some Indian institutions are given as ranges because published accounts differ slightly.

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